Guidelines for Meshing Pump (stage geometry) for CFD Analysis

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Index

1. Purpose 3
2. Scope 3
3. Abbreviations 3
4. General procedure: description 3
   1. **Numerical Discretization techniques: FVM** 4
   2. **Element types of cell mesh** 5
   3. **Typology of mesh: structured and unstructured grid** 6
   4. **Number of grid elements and simulation time.** 7
   5. **Extension Inlet and Outlet Mesh** 8
   6. **Interface of mesh** 9
   7. **Boundary Layer: Modeling Flow Near the Wall and Height of first cell** 9
   8. **Boundary Layer: Grow rate, boundary layer.** 11
   9. **LE TE blade and splitter mesh.** 12
   10. **Leakage Mesh** 14
   11. **Mesh quality.** 15
5. Acceptance criteria 18
6. References 20

Annex A: Creation of mesh pump using Meshing and ICEM (Ansys®) software 21

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1. Purpose

The document describes a guideline to realize the mesh of a pump (single o multistage) with a quality level sufficient to adequately capture fluid-dynamic phenomena minimizing the number of elements of the computational grid.

A specific example using MESHING and ICEM of ANSYS environment is described in the Annexes.

2. Scope

This procedure is applicable to any centrifugal pumps: Overhung, Between-bearings, Vertical Suspended; also, all types of arrangements, single or multistage, suction simple o double entry, volute discharge or diffuser discharge.

The procedure applies to the Nova Milanese and La Spezia plants.

3. Abbreviations

LE		*leading edge of blades impeller*
TE		*trailing edge of blades impeller*
PS		*pressure side of blades impeller*
SS		*suction side of blades impeller*
D2	*[mm or in]*	*maximum external diameter of the impeller*
D1	*[mm or in]*	*impeller eye diameter*
Dh	*[mm or in]*	*shaft impeller diameter*
Nq or Ns		*specific speed of the pump*
Eta	*[%]*	*efficiency*
Q	*[m^3/h]*	*flowrate*
H	*[m]*	*head*
CFD		*Computational fluid Dynamics*

4. General procedure: description

The guideline provides general instructions to design a computational grid of pump starting from a cad model (in general assembly model) that will be used for CFD analysis. Based on the type of arrangements pump CAD extracted and prepared, in general the components that make up the assembly of pump are the following:

- 3D model of suction
- curves/2D/3D model of impeller/bowl/simple diffuser

COPY COMPLIANT to the original document resident on the computer system at the time of printing

- 3D model of volute/complex diffuser
- 3D model of crossover
- 2D/3D model of leakage
- 3D drum and flushing lines
- 3D extension numerical mesh

It is possible to use different types of grid generators for mesh the components of pump; depending on the available type of cad, the following different approaches will be adopted to create the mesh.

The impeller mesh guideline is not covered in this document, as it is already discussed extensively in **SOP-490**.

A general discussion is presented below, with the aim to provide the user with the necessary information and recommended criteria for meshing the pump using general purpose tools.

1. Numerical Discretization techniques: FVM

Analytical solutions of the Navier-Stokes equations are able to solve only the simplest flows under ideal conditions. To obtain solutions for real flows, a numerical approach must be adopted whereby the equations are replaced by algebraic approximations which may be solved using a numerical method.

The techniques of Finite Volume Method is the "classical" or standard approach used most often in commercial software and research codes. T

he governing equations are solved on discrete control volumes FVM (Figure1) recasts the PDE's of the N-S equation in the conservative form and then discretize this equation.

This guarantees the conservation of fluxes through a particular control volume.

Though the overall solution will be conservative in nature there is no guarantee that it is the actual solution.

Moreover, this method is sensitive to distorted elements which can prevent convergence if such elements are in critical flow regions.

The following guidelines provide a series of criteria and settings developed over the years through a persistent series of analyzes and comparisons that guarantee an adequate degree of reliability of the results obtained compared to the experimental data measured in the field.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

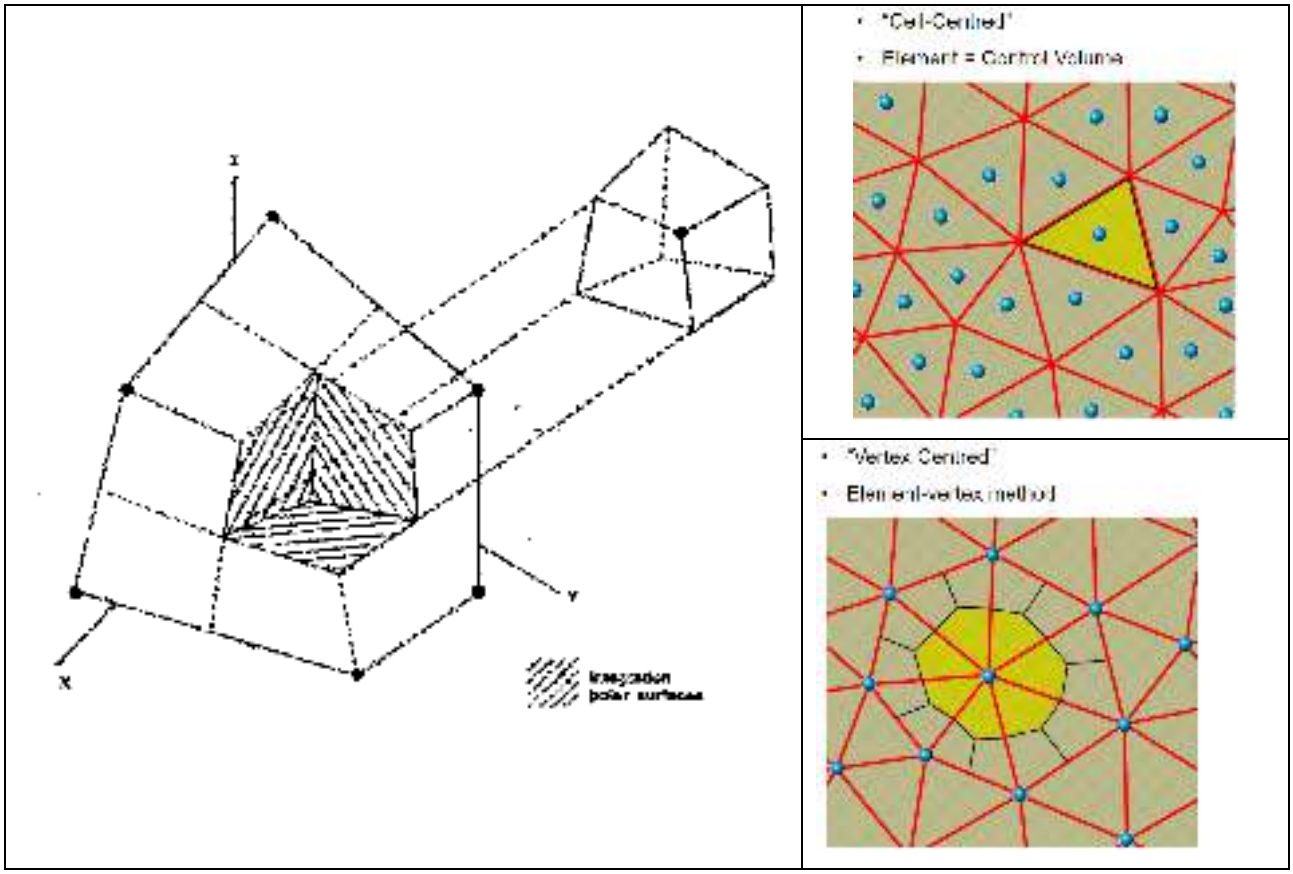


Figure 1 : FVM, method Cell-Centred and method Vertex-Centred

2. Element types of cell mesh

Figure2 shows different kinds of mesh element, the appropriate choice and placement of these elements that discretize the fluid domain depends on which fluid dynamic phenomenon you want to model and with what degree of accuracy.

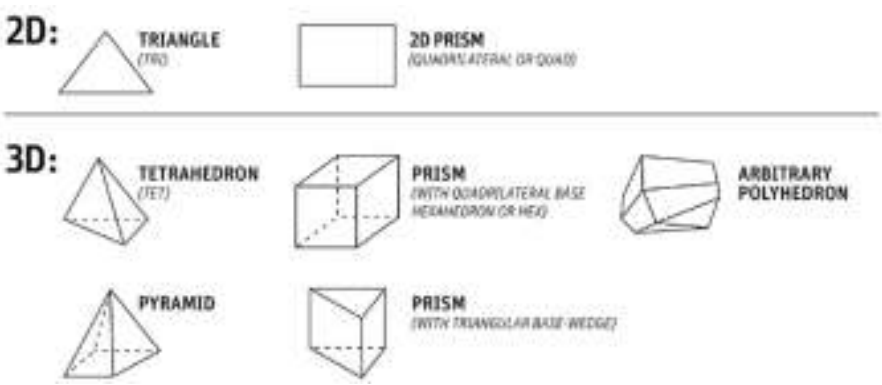


Figure 2 : type mesh element.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Prismatic elements are the best in terms of quality, but they cannot always be used due to the irregular shapes of the geometry, however it is recommended to always use them to simulate the boundary layer.

3. Typology of mesh: structured and unstructured grid

Mesh elements can be generated according to two different types of logic (Figure3):

Non structured:

Any agglomeration of elementary elements organized or not. Most of the meshes in this category are hybrid meshes using tetrahedrons and prisms.

Structured:

An organized agglomeration of hexahedrons.

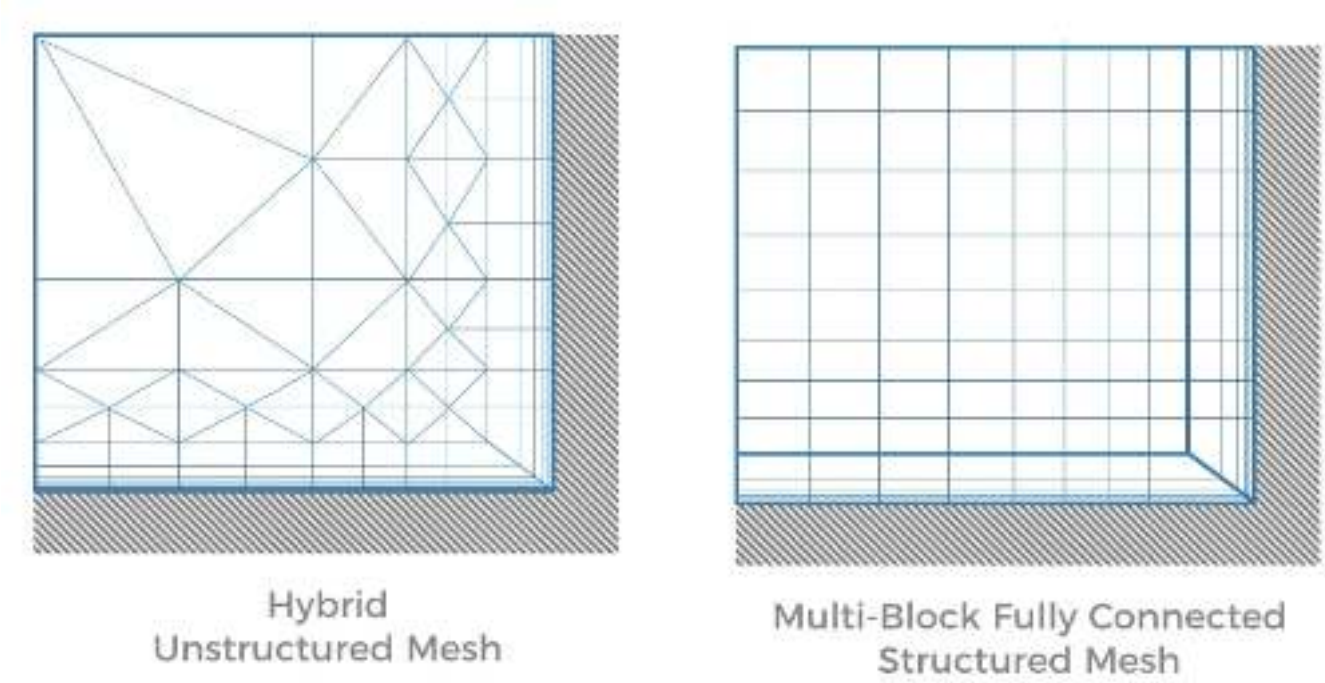


Figure 3 : structured and unstructured grid.

The choice between the two types of mesh depends on the complexity of the geometry, the accuracy of the numerical result we want to obtain and the type of mesh tool available.

Although hexahedral elements are best at resolve and transmitting information between adjacent cells, they are only applicable for simple geometries; for complex shaped geometries it is necessary to switch to an unstructured grid.

In general, if you have "2D leakage", “3D suction pipe” and “3D extension numerical mesh” available, it is possible to create a Structured Mesh (for example using the ANSYS ICEM tool), while

COPY COMPLIANT to the original document resident on the computer system at the time of printing

with the other geometries "3D model" it is easier to create a Hybrid Unstructured Mesh with a generic tool (for example ANSYS MESHING tool).

4. Number of grid elements and simulation time.

The final number of mesh elements varies from case to case and depends on the degree of refinement you want to give, since the mesh must be refined in zones with high velocity gradients, so near the vane LE - TE and near wall.

Sometimes we are forced to increase the number of elements when the quality of the mesh is not good, in this phase it is important to distribute the mesh elements appropriately because even if it is true that a greater number leads to a better quality of the solution but on the other hand it requires more simulation time.

The following table shows typical ranges of number of total elements needed to have a good quality mesh:

Type of component	Typical range number of element	Qualitative refinement scale (1 low - 4 high)	Minimum number of elements in the cross section (without boundary layer)
3D model of suction	500 0000 – 2 000 000	2	15
3D model of volute/complex diffuser	1 000 000 – 3 000 000	4	8
3D model of crossover	500 0000 – 1 000 000	3	10
3D model of leakage	1 000 000 – 1 500 000	4	15
3D drum and flushing lines	100 000 – 500 000	2	5
3D extension numerical mesh	100 000 – 200 000	1	15

The table also shows the minimum number of elements that must be generated in a cross-section, not considering the number of elements imposed in the boundary layer; furthermore, Figure 4 shows in a qualitative way an example of density and distribution of the elements in a cross-section of a generic multistage pump.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

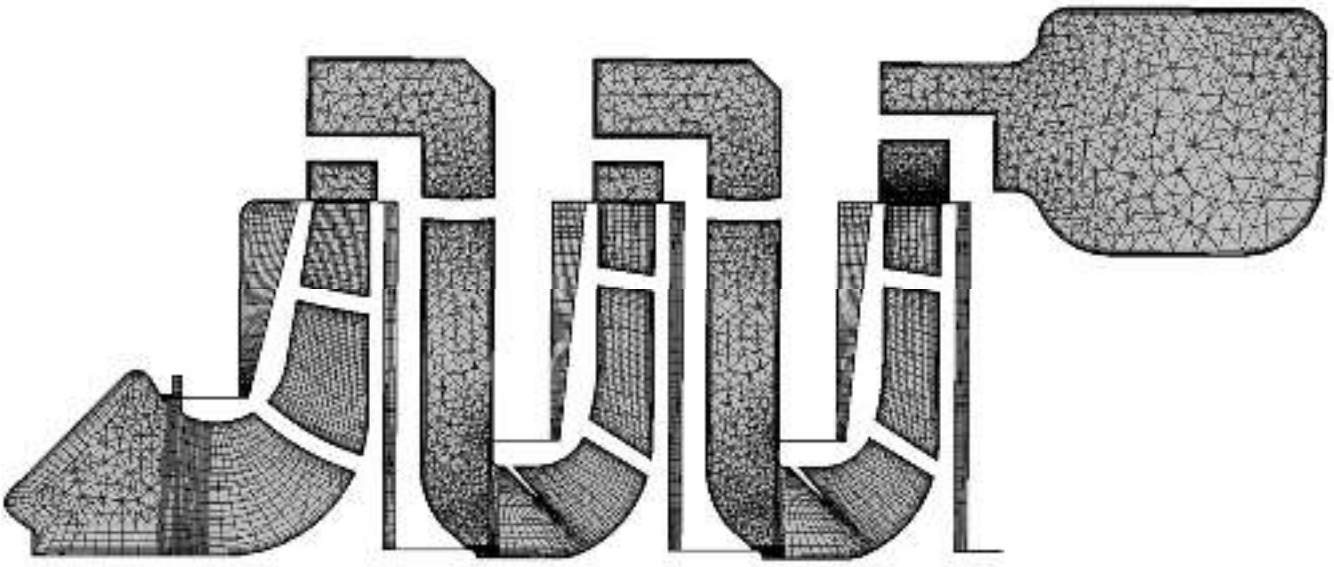


Figure 4 : boundary layer model.

5. Extension Inlet and Outlet Mesh

The inlet and outlet of the pump mesh are regions for which a boundary condition is assigned; it is common practice to extend these two regions with numerical domains so that the boundary conditions do not excessively disturb the regions inside the pump where the fluid develops.

The company standard requires an extension of the suction domain of 2 times the diameter of the suction pipe flange and the discharge domain extended by 4 times the diameter of the discharge pipe flange.

	Extension domain suction	Extension domain discharge
Diameter flanges	2D	4D

Very often for pumps with a high number of stages only a model of part of the pump is created, if this ends with the diffuser component, also in this case an extended 4D annular duct must be applied considering the presence of the pump shaft.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

6. Interface of mesh

To have conformity in the interfaces between the stator and rotor meshes or stator and stator mesh, they must have approximately the same number of elements and approximately the same distribution.

If the cells are particularly distorted in terms of orthogonality and aspect ratio, it is possible that an incorrect transmission of information between the two meshes may occur.

7. Boundary Layer: Modeling Flow Near the Wall and Height of first cell

Near the solid wall, the modeling of the fluid boundary layer requires particular attention, since in this region strong gradients of the fluid dynamic variables calculated by the CFD solver occur.

Experiments and mathematical analyzes have shown that the region near the wall can be divided into two layers:

- in the innermost layer, close to the wall, the so-called "viscous sublayer", the flow is almost laminar and the (molecular) viscosity plays a dominant role in the momentum and heat transfer.
- in the layer furthest from the wall, in the logarithmic layer, turbulence dominates the mixing process.

The Figure5 below illustrates these subdivisions of the near-wall region and an example of discretization of boundary layer.

Note as a fine grid resolution is required to accurately resolve the gradient of velocity.

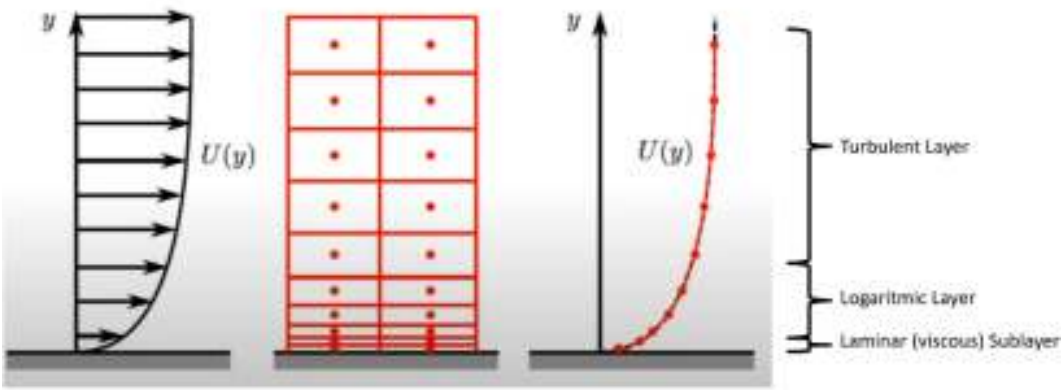


Figure 5 : boundary layer model.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

The boundary layer can be either modeled using "wall functions" (empirical functions obtained from experiments and DNS simulation that are fitted to the observed behavior close to the wall) or simulated by imposing the smallest possible height of the first cell and an adequate growth ratio.

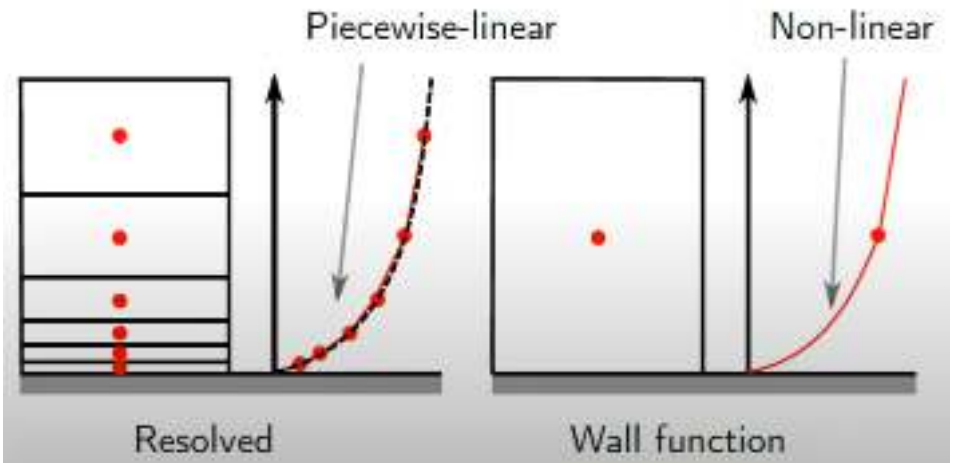


Figure 6 : boundary layer model, resolved and wall function.

Since different stationary/unstationary velocity fields arise near the wall in the fluid domain, it is not possible at present to adapt the height from the first cell to the various needs, we are therefore forced to a compromise value.

	Height of first cell [μm]	
	Wall function	Low Reynolds method
Smooth wall	200-300	1
Wall with roughness	5*Ra	/

The following Figure7 shows the relative height value of the first cell depending on the roughness Ra (Averaged Roughness) in American and Metric system.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Approximate Surface Roughness Conversion Chart

Roughness Grade Numbers	American System		Metric System		Height first cell mesh (µm)
	Ra (µin)	RMS (µin)	Ra (µm)	RMS (µm)	
N12	2000	2200	50	55	64
N11	1000	1100	25	27.5	32
N10	500	550	12.5	13.75	16
N9	250	275	6.3	9.13	8
N8	125	137.5	3.2	3.52	4
N7	63	69.3	1.6	1.76	2
N6	32	35.2	0.8	0.88	1
N5	16	17.6	0.4	0.44	1
N4	8	8.8	0.2	0.22	1
N3	4	4.4	0.1	0.11	1
N2	2	2.2	0.05	0.055	1
N1	1	1.1	0.025	0.035	1

Figure 7 : heigh first cell mesh with Ra.

For further details consult [2]

8. Boundary Layer: Grow rate, boundary layer.

Once the method that defines the thickness of the first cell has been chosen, which as mentioned in the previous paragraph we conventionally assume to be equal to:

Height of first cell for “Wall functions”: 200 micrometers.

Height of first cell for “Low Reynolds method “: 1 micrometer

It is necessary to define the growth ratio of the cell height G and the number of cells N.

Two good rules are listed below (Figure8):

1) the boundary layer thickness δ_{99} must be contained within the inflation layers.

2) Increase the N number of layers keeping recommended G in the range 1.05<G<1.4.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

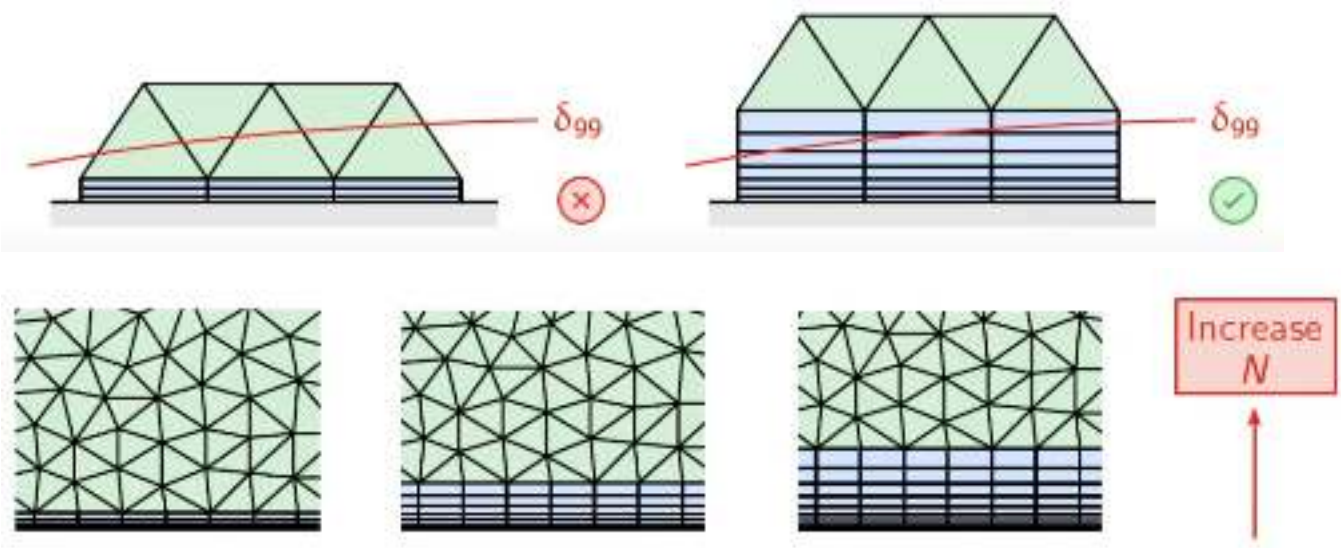


Figure 8 : Good rules for G and N

Since the thickness of the boundary layer is not known regardless, also in this case the following conventional values are assumed:

N= 8-16

G=1.4

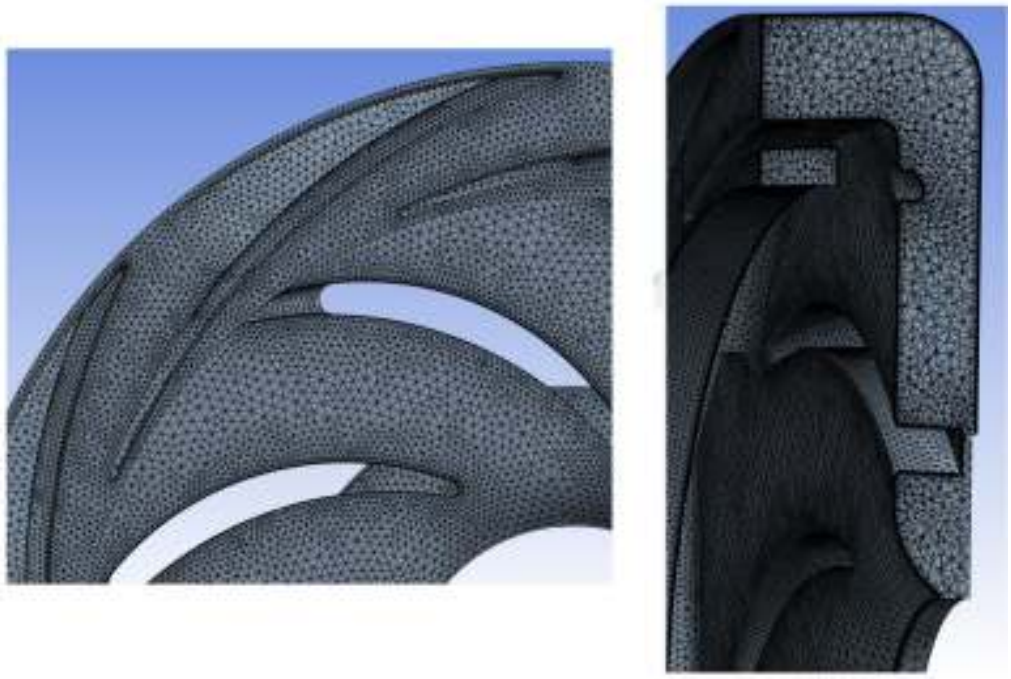
Figure 9 : example elements in the internal domain.

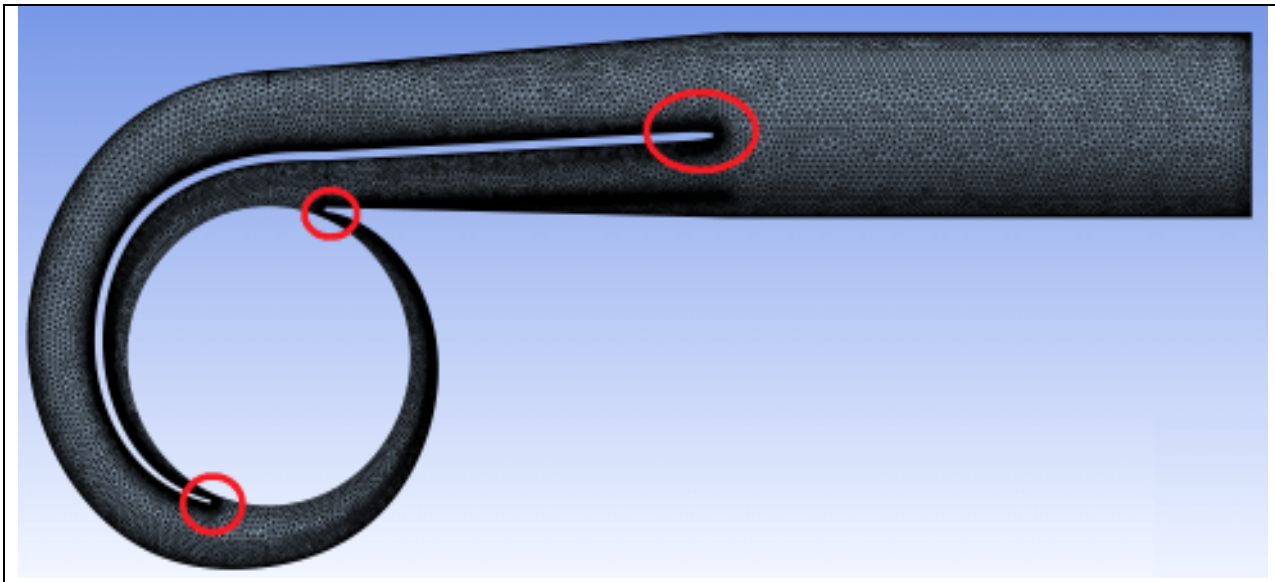
9. LE TE blade and splitter mesh.

The fluid region close to the LE is critical as it is the region of incidence of the flow on the blade/splitter, therefore it must have an adequate degree of discretization, generally 8-20 elements arranged on the LE are sufficient to follow its curvature well.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

The fluid region close to the TE is also critical since downstream of it a turbulent wake is created whose size and intensity are mainly linked to the thickness of the blade in this section, generally 8-20 elements arranged on the TE are sufficient to follow its shape well (Figure10).





COPY COMPLIANT to the original document resident on the computer system at the time of printing

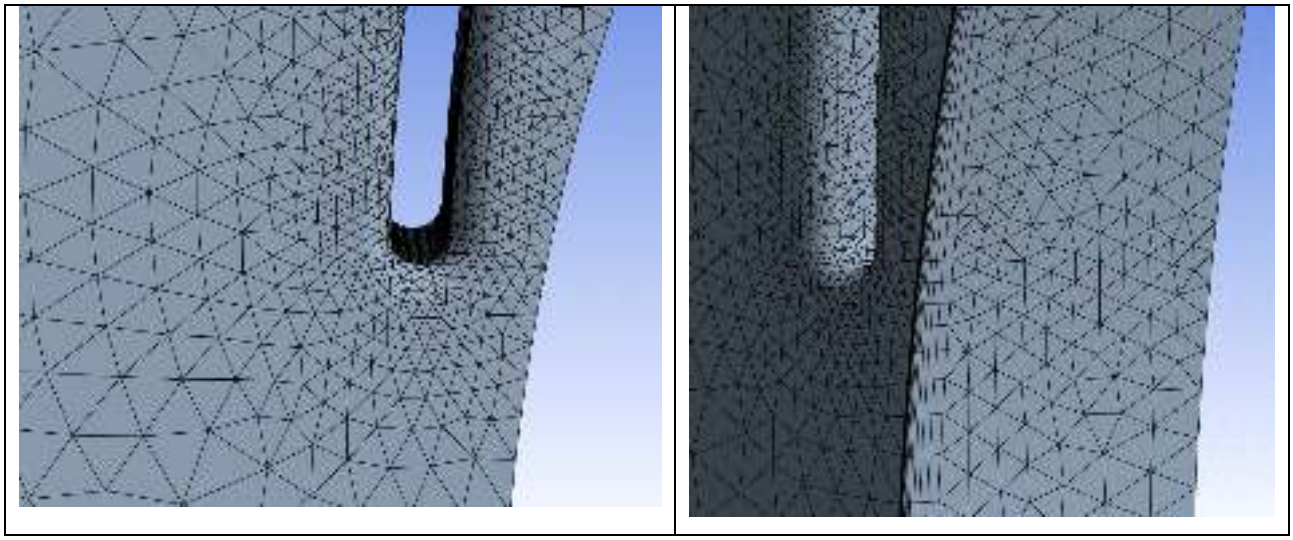


Figure 10 : example refinement diffuser and volute mesh.

10. Leakage Mesh

Usually, because the leakage is axisymmetric, it is preferable to create a structured mesh; regardless of this the leakage mesh must have:

- a thickness of the first cell of the limited layer of 1 micrometer
- a number of boundary layer cells equal to N=10
- a growth ratio G=1.4
- a number of cells in the leakage between the wear rings (without taking into account the boundary layer) equal to 6 in the radial direction (see Figure11).

COPY COMPLIANT to the original document resident on the computer system at the time of printing

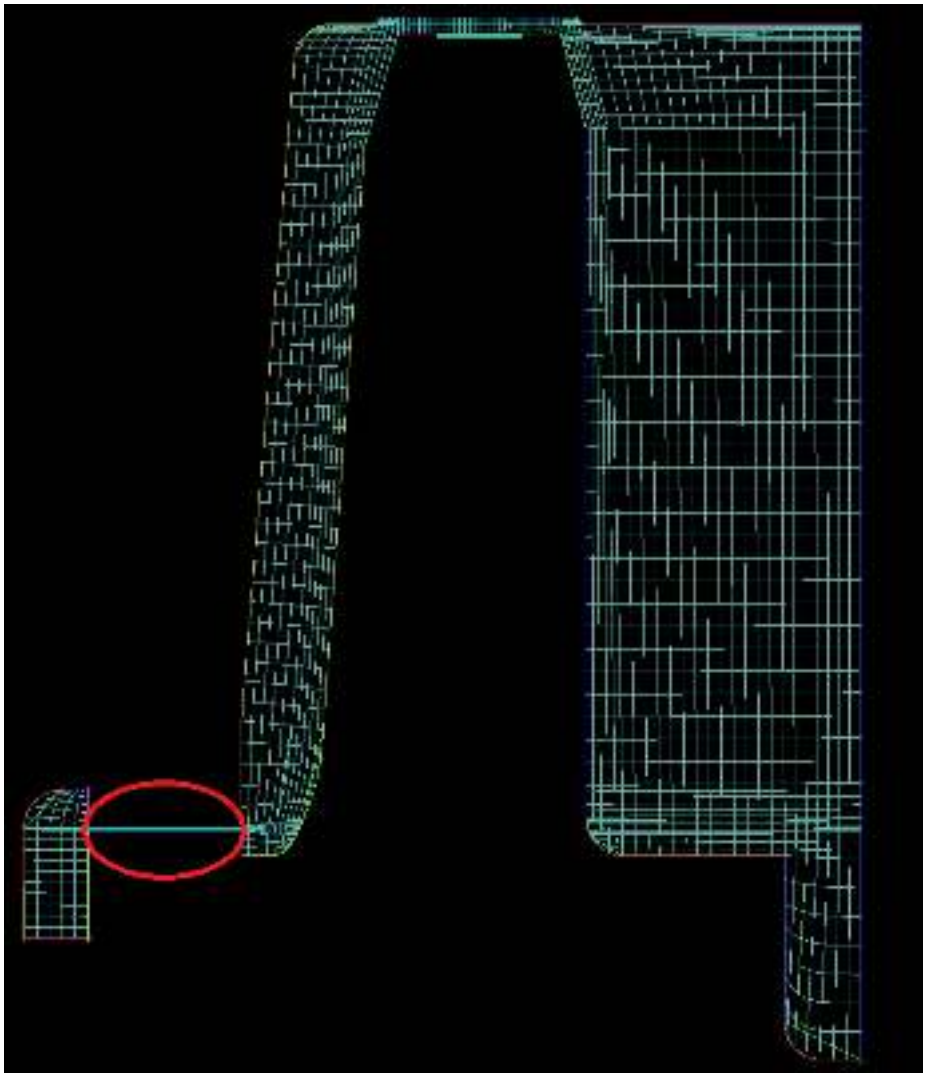


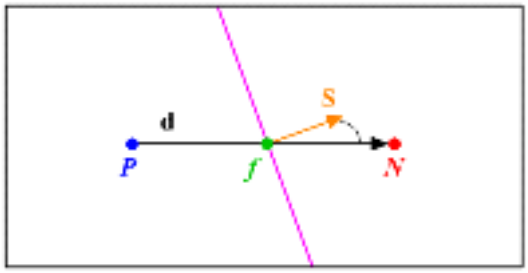
Figure 11 : example refinement

11. Mesh quality.

The solution of a simulation with FVM strongly depends on the quality of the mesh and the robustness of the solver, so that the results can be considered a valid approximation of the real physical phenomenon the simulation must respect the three fundamental concepts of: consistency, stability and convergence [1].

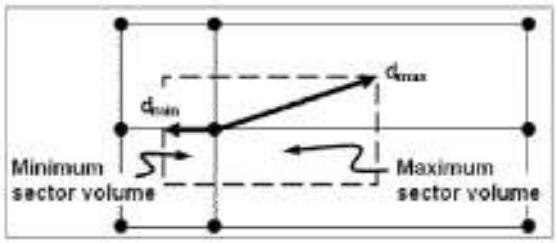
In the literature there are various criteria for evaluating the quality of the mesh, but the main ones are reported and defined in the following Figure12:

COPY COMPLIANT to the original document resident on the computer system at the time of printing



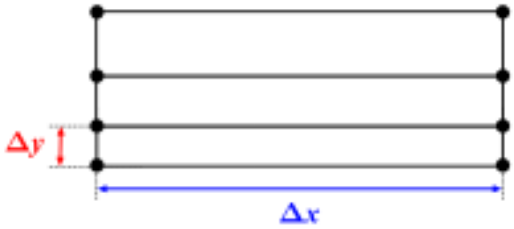
Mesh Orthogonality

Is the angular deviation of the vector S (normal at the diagonal and located at the face center f) and vector d (connecting the two-cell center P and N)



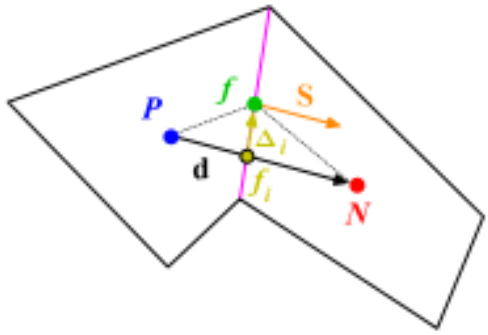
Mesh Expansion Factor

is the ratio of the volume of the larges and smallest sector volumes in element



Mesh Aspect Ratio

Is the ratio of the longest edge length to the shortest edge length of element



Mesh Skewness

Is the deviation of the vector d that connects the two cell P and N, from the face center f.

the deviation vector is represented with Δ and fi is the point where the vector d intersects the face f.

Value of Skewness	Cell Quality
1	degenerate
0.9 — <1	bad (sliver)
0.75 — 0.9	poor
0.5 — 0.75	fair
0.25 — 0.5	good
>0 — 0.25	excellent
0	equilateral

Figure 12 : Criteria for Mesh Quality

COPY COMPLIANT to the original document resident on the computer system at the time of printing

The acceptability ranges for the different defined criteria are listed below:

Criteria suggested by Gulich [1]

To be used if you do not have references on the quality criteria suggested in the mesh tool and solver.

Mesh Orthogonality: the grid lines should be as orthogonal as possible to the walls and within the calculation domain. The elements must not have angles less than 20° or greater than 160° (angles less than 20° compromise not only precision but also convergence).

Mesh Expansion Factor: the size of the elements should not change sharply. The expansion rate of the cells should not change from one cell to the next by more than a factor of 1.1 near walls and 1.5 to 2 in the bulk flow domain; or the cell volume should not increase more than by a factor of 1.5.

Mesh Aspect Ratio: the aspect ratio of the elements should not exceed the factor 10 near walls and 5 in the bulk flow regions. It is particularly important to fulfill these requirements in zones with strong velocity gradients.

Mesh Skewness: the skewness should have at least good or better quality.

No intersections of grid lines (no negative volumes) are allowed.

Internal criteria adopted by the Company

The following criteria aim to exclude that type of mesh which, once created, can create problems in the solution of the calculation, leading to failure.

To be accepted, the mesh must possess these fundamental requirements:

- all mesh elements must have a size greater than 1 micrometer.
- when generating the mesh, elements with negative volume must not be created.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

5. Acceptance criteria

This procedure is applicable for any geometry of suction, cross-over, volute and diffuser (Table 1, Table 2, Table 3, Table 4).

In the following tables are described the different typologies of static component which constitute a centrifugal pump:

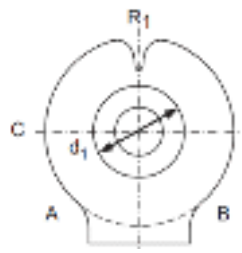
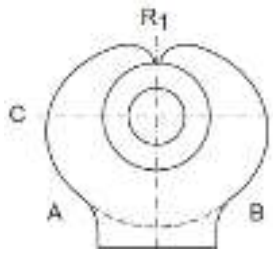
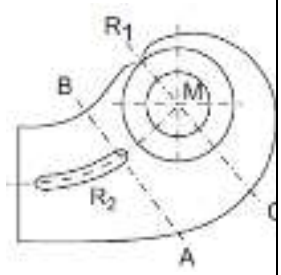
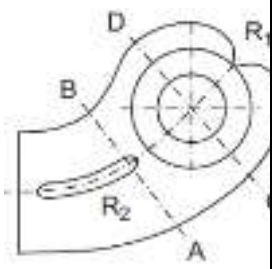
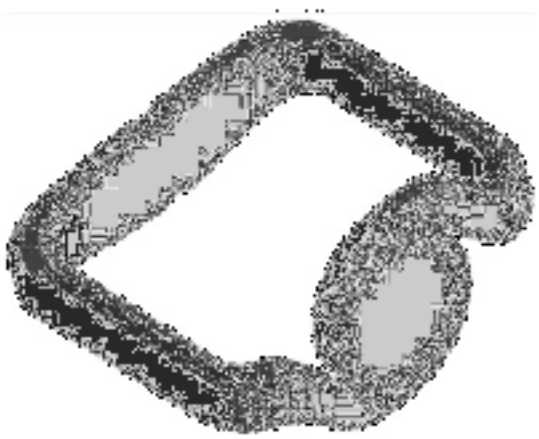
Type suction casings			
Barrel pump (radial split)		Double suction (axial split)	
anulus	Spiral shaped	asymmetrical	quasi-symmetrical
			

Table 1 : Type of geometry suction [1]

Short and long cross-over for multistage	
Short Xover	Long Xover
	

COPY COMPLIANT to the original document resident on the computer system at the time of printing

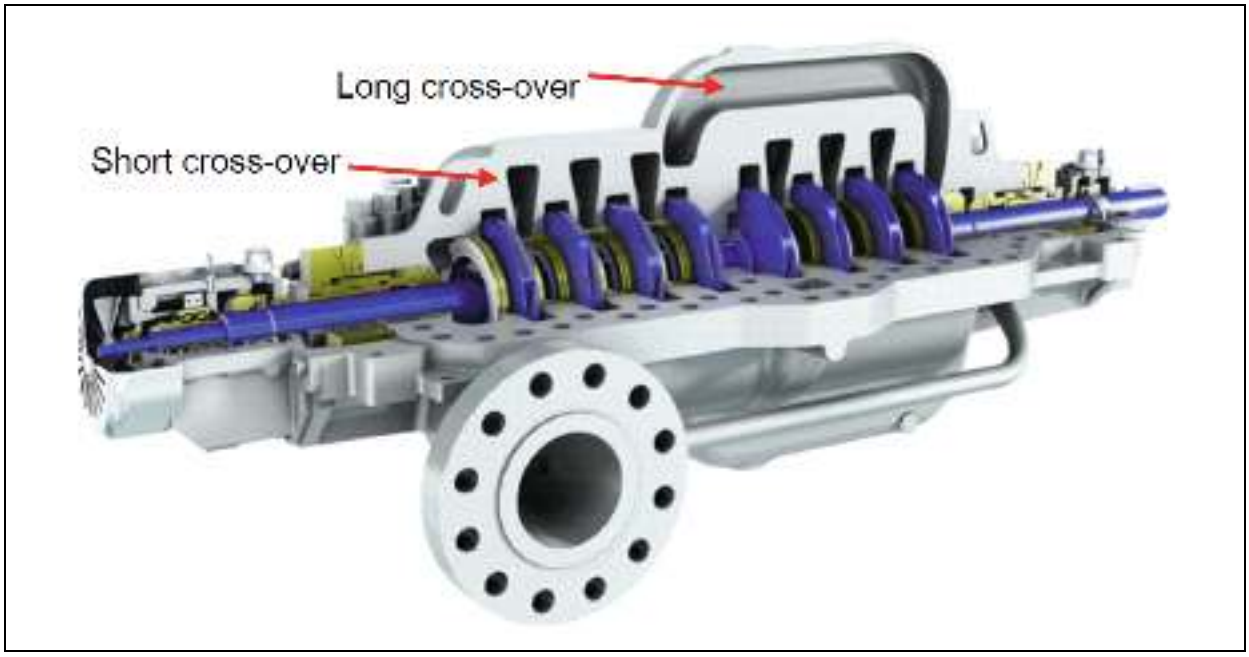


Table 2 : Type of geometry cross-over [1]

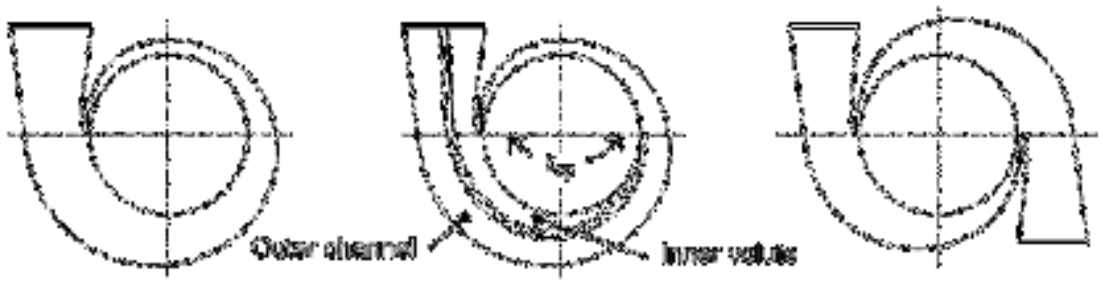
Type volute casings		
Single volute	Double volute	Twin volute
		

Table 3 : Type of geometry volute [1]

COPY COMPLIANT to the original document resident on the computer system at the time of printing

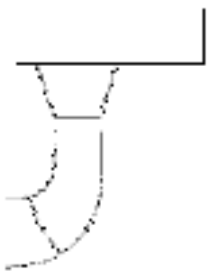
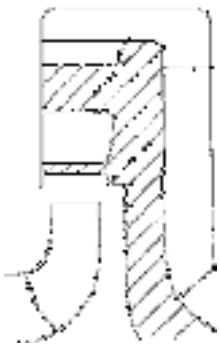
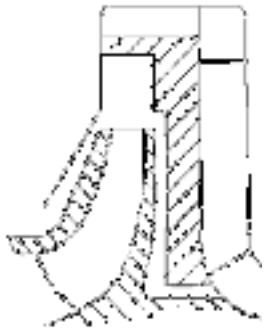
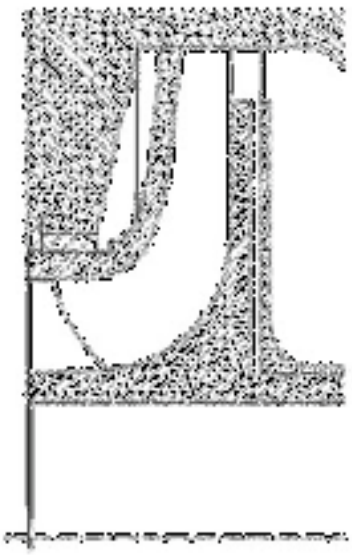
Type diffusers casings (diffusers and return vanes separated)			
Diffuser with volute	Diffuser and annular casing as last stage	Overflow from diffuser to return vanes through vaneless annuls	Lateral overflow from diffuser to return vanes
			
Type diffusers casings (diffusers and return vanes continue)			
			

Table 4 : Type of geometry diffuser [1]

By the type of pump will be necessary to work on different composition of this part of pump.

6. References

[1] Gulich "Centrifugal Pumps" ,4th.ed.

[2] Ansys User Manual

[3] https://www.cfd-online.com/

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Annex A: Creation of mesh pump using Meshing and ICEM (Ansys®) software

The best practice is to follow every step described in this chapter in the proper order with the aim to reduce the number of operations.

1. Introduction

Before starting the meshing phase, several CAD preparation steps are necessary; it can be summarized in the following three phases:

- assembly
- extraction of fluid domains
- division of different fluid domains: the position of the mesh interfaces is identified (also with the help of a 2D drawing).

The reason why it is necessary to divide the fluid domains is multiple, they are the following:

- the fluid domains are both stator and rotor, a physical separation between the stator and rotor domain is therefore necessary.
- it is convenient to create different types of meshes with different meshing tools because each component has its own geometric complexity and a different mesh definition level
- subdividing and distinguishing the different pump components allows you to estimate its performance independently and allows you to carry out a possible flexible and targeted redesign activity of the individual component.

Because is not possible resume all type of pump in only guideline, will be reported an example of meshing of model multistage (3 stages) pump (BB4 Nq20).

COPY COMPLIANT to the original document resident on the computer system at the time of printing

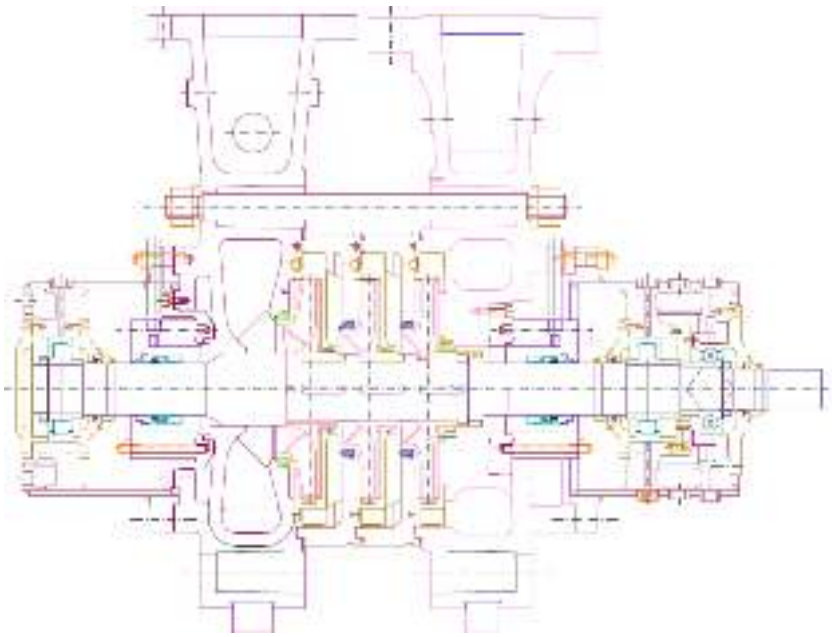
Sectional drawing of prototype pump multistage (3 stages)


Table 5 : 2D drawing BB4 Nq20 3 stages.

The following tables show the different fluid components that make up the pump in question:

Suction	Diffuser of first stage	Diffuser of second stage	Diffuser of third stage	Discharge

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Table 6 : static component of pump

From the assembly cad 3D is necessary to have drawn the leakage:

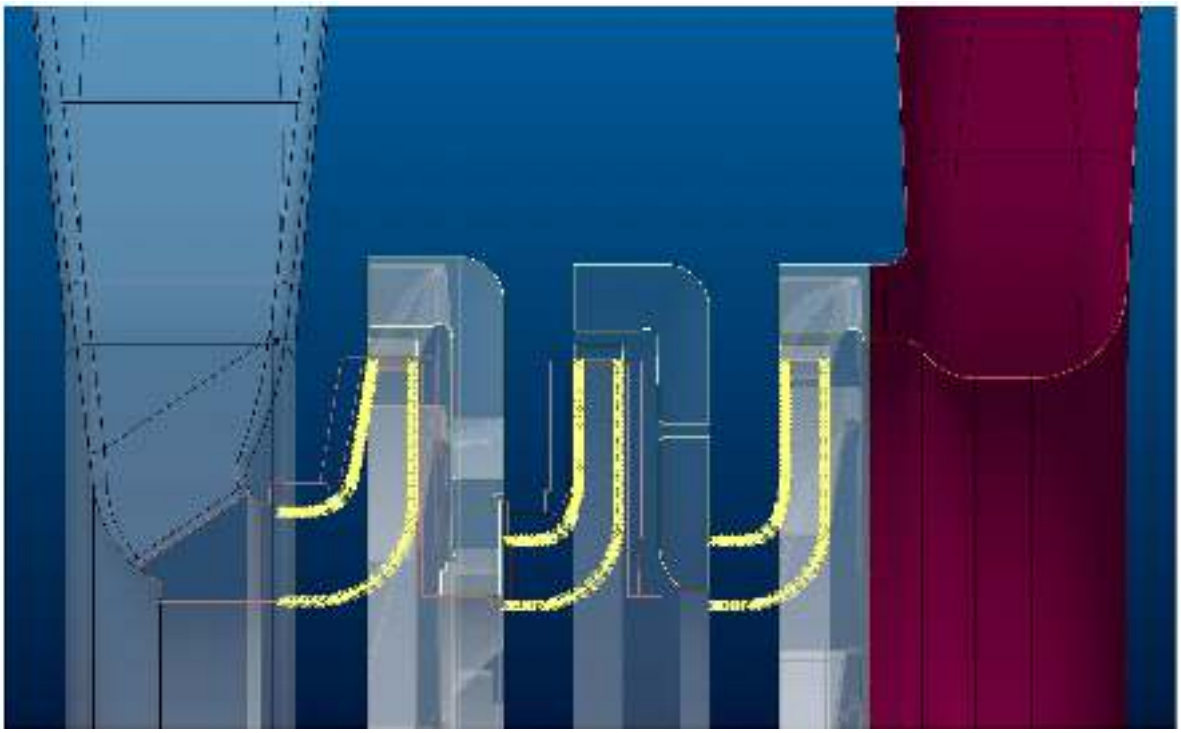
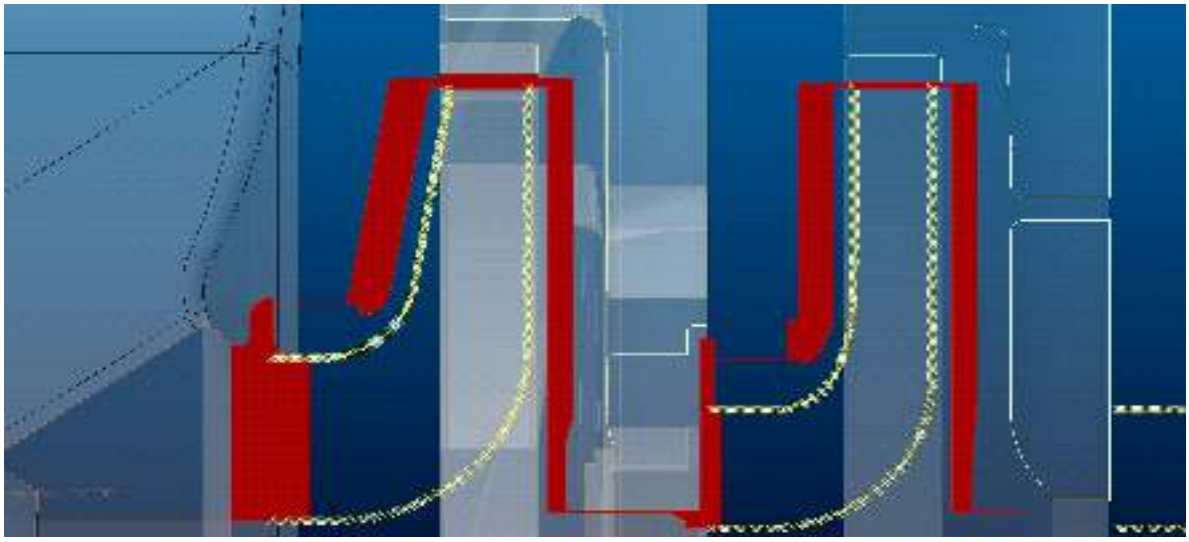
Assembly cad 3D

With red color the two leakages obtained from the two sketches (first stage and successive stages)


Table 7 : assembly and leakage of pump

The input files are following:

COPY COMPLIANT to the original document resident on the computer system at the time of printing

Type component	geometry	File extension	Type tool of meshing
suction	3D	.iges/.step	ANSYS MESHING
leakage	2D	.iges	ICEM
Diffuser1/2/3	3D	.iges/.step	ANSYS MESHING
discharge	3D	.iges/.step	ANSYS MESHING

Table 8 : input geometry file for mesh

2. General Procedure – Description

Because the procedure used for meshing the component of suction, cross-over, volute and diffuser is about the same, for simplicity will be described as to mesh a diffuser using ANSYS MESHING and as to mesh a leakage using ICEM.

MESHING DIFFUSER

2.1. Open the environment ANSYS WORKBENCH, drag and drop the two block “MESH” and “CFX” in the working area and link them as in Figure 13, then click “Import Geometry” for import the file diffuser.iges inside the first block in the section geometry.

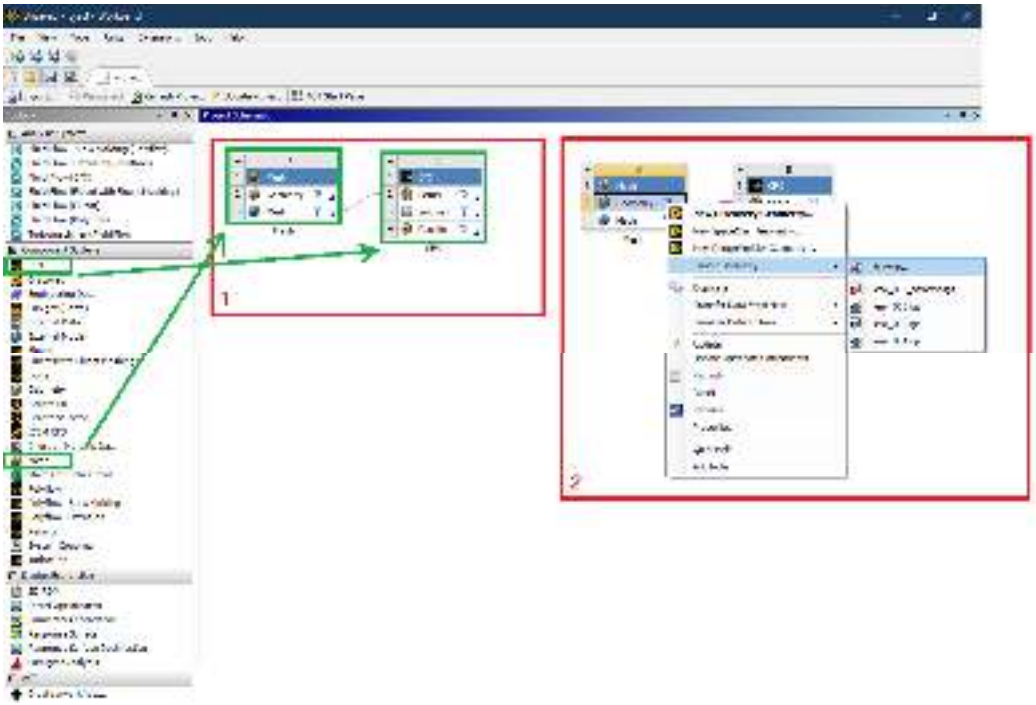


Figure 13 : working area of ANSYS WORKBENCH

COPY COMPLIANT to the original document resident on the computer system at the time of printing

2.2. Click edit mesh for create a new mesh into the domain imported; into the environment "MESH" compile the fields in "Details of Mesh" (Figure 14) as follows:

- ***Physics Preference**: CFD*
- ***Solver Preference**: CFX*
- ***Element Size**: 0.008m* ; this is the mainly dimension of cell without considering refinements and boundary layers, this number is a first attempt value, indeed it depend of main dimension of domain and the degree of resolution accuracy you want to have, so it is arbitrary.
- ***Defeature Size**: 0,001m* ; this command is a filter that allowed of reduce the resolution of geometry, in other words the value inserted is the minimum geometrical value that the tool of meshing see into the cad of geometry. This filter allowed so not meshing where there are micro-edge or micro-angle. This value should be as low as possible so that the mesh created (discretization of the domain) is more similar to the real geometry, so it is arbitrary but must be obviously minor of "Element Size".

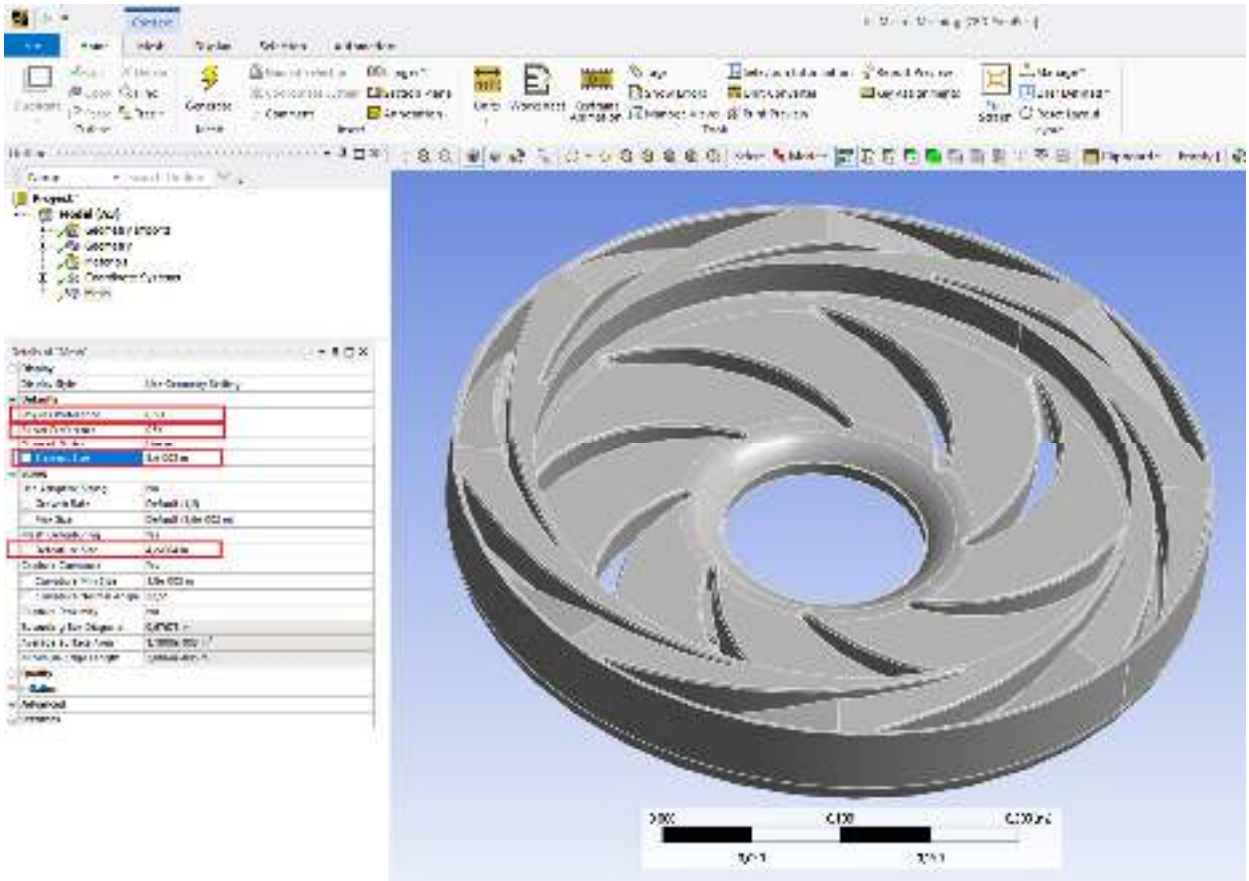


Figure 14 : Set details of mesh

All the other fields let them as in default.

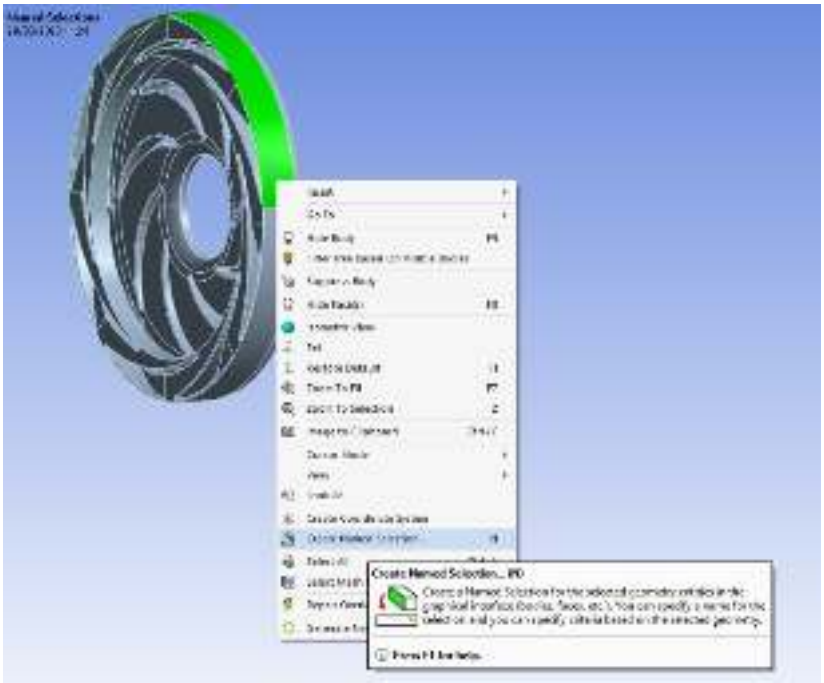
COPY COMPLIANT to the original document resident on the computer system at the time of printing

2.3. Create name selection of all geometry's faces, as follows in Figure 15:

Use the selection tools for select the faces and attribute them the appropriate name.

For the diffuser it is necessary to name 3 patches:

- **leak2diff**: it is the interface between the leakage domain and the diffuser domain of the same stage.
- **wall_diff**: these are all the surfaces of the solid wall of the diffuser
- **diff2leak**: is the interface between the diffuser domain and the leakage domain of the next stage.



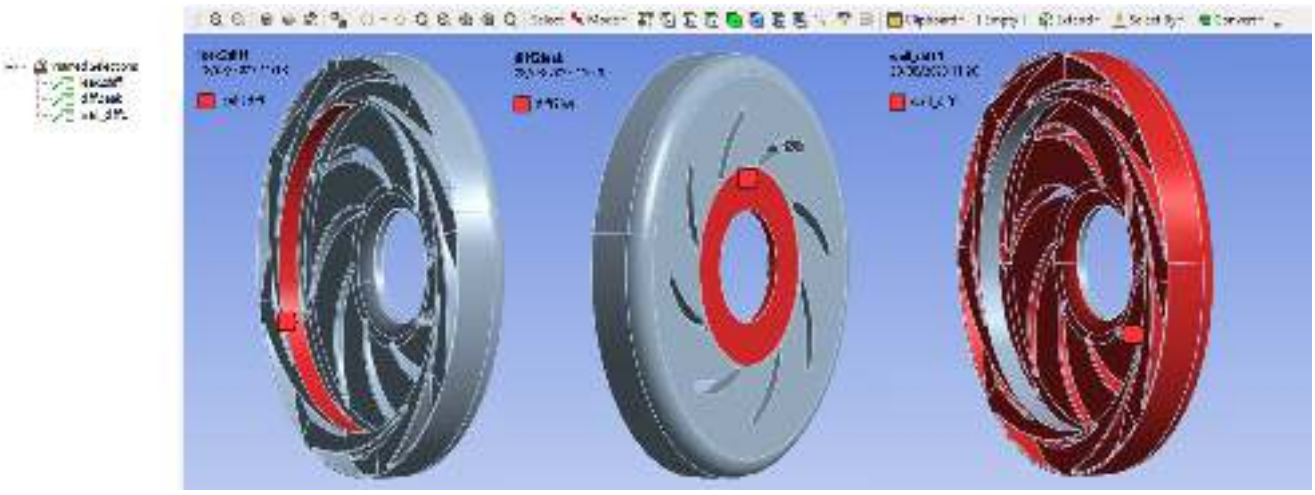


Figure 15: Creation of name patch

COPY COMPLIANT to the original document resident on the computer system at the time of printing

This step is useful for recognize later inside the environment CFX the different patch where to assign the boundary conditions.

2.4. Insert a specific method of mesh.

Usually 2 meshing methods are used:

- **INFLATION**: allows you to create the boundary layer on the solid wall of the component.
- **SIZING**: allows you to create localized mesh refinements on edges and faces (is useful in CFD for refinement the mesh in specific region where is necessary resolve the flow with better degree of accuracy, for example near the leading edge of the blade, on the interface's surfaces, on edge where the inflation cannot go around an edge of the geometry and in other region (edges, surfaces, volume region) where the quality mesh is lower of the recommended values of quality mesh).

The settings for the "Inflation" method are as follows (see Figure 16):

- **Geometry selection:** select the geometry and all surfaces of wall.
- **First Layer Thickness**: in function of roughness to simulate, refer to the table in Figure6.
- **Maximum Layers**: range 8-16, refer to the Cap.8.
- **Growth Rate**: range1.1-1.4, refer to the Cap.8.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

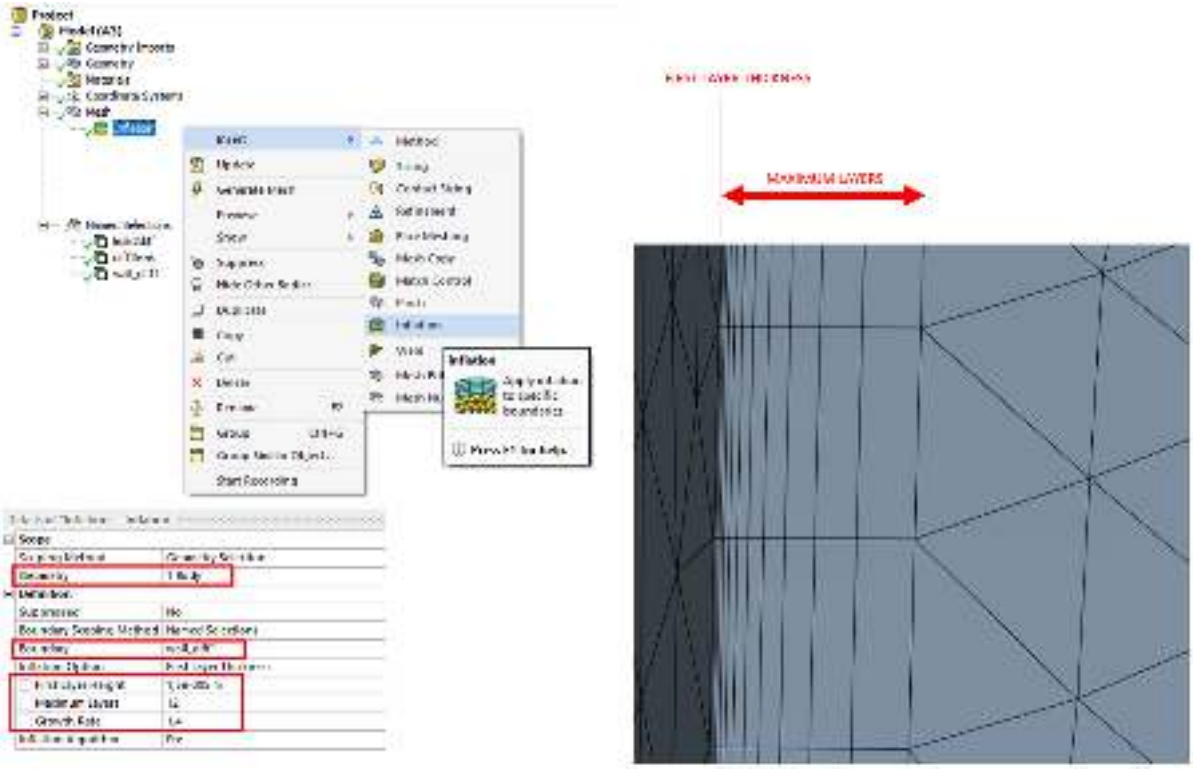


Figure 16: Insert inflation for boundary layer.

The settings for the "Sizing" method are as follows (see Figure 17):

- **Geometry selection:** select the Edge o Faces for the refinement of local mesh
- **Element size/Number of element**: select the element size or number of elements to attach to the specific edge or face.
- **Growth Rate**: 1.1

COPY COMPLIANT to the original document resident on the computer system at the time of printing

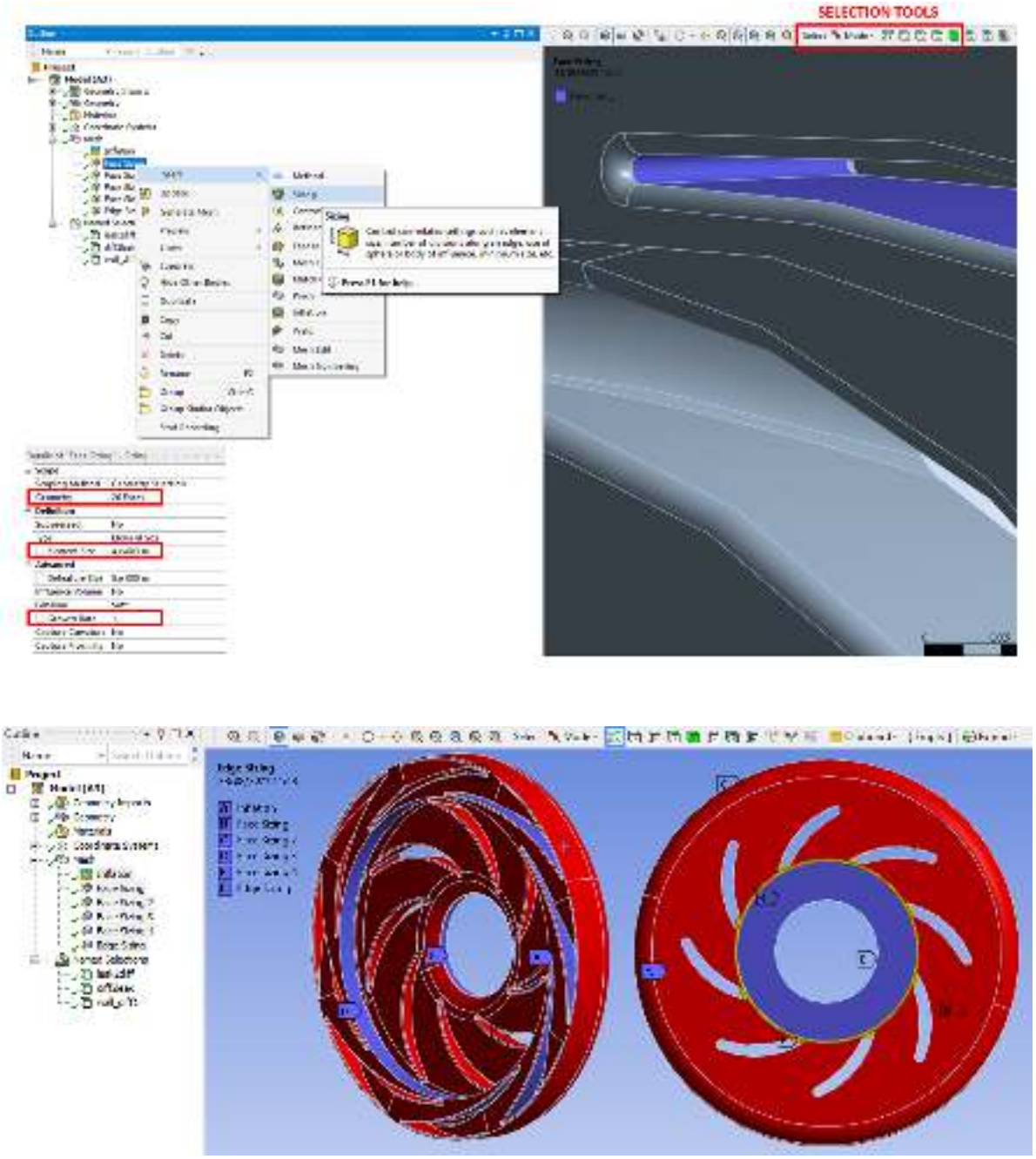


Figure 17: edge/face sizing for refinement mesh in particular regions.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

2.5. Click the button "Generate" for start the meshing process, in Figure 18 an example of mesh done correctly.

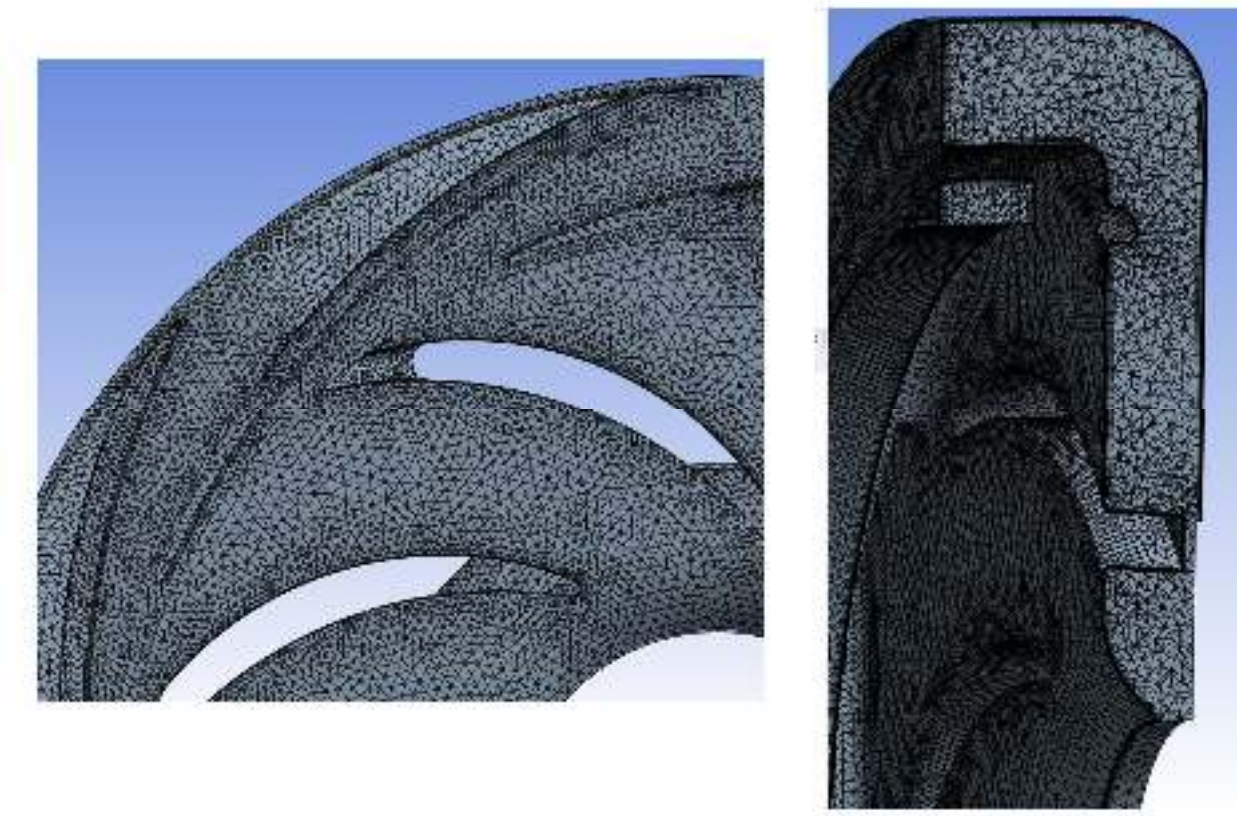


Figure 18: example of mesh done correctly

2.6. <u>Check quality mesh</u>

In the Quality section use the "**Element Quality**" criteria for evaluating the quality of the mesh.

The practice adopted by the company is that the mesh is sufficiently accurate if the "Prism" elements (Pyr5 in Figure 19) have all exceeded a value of 0.13.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

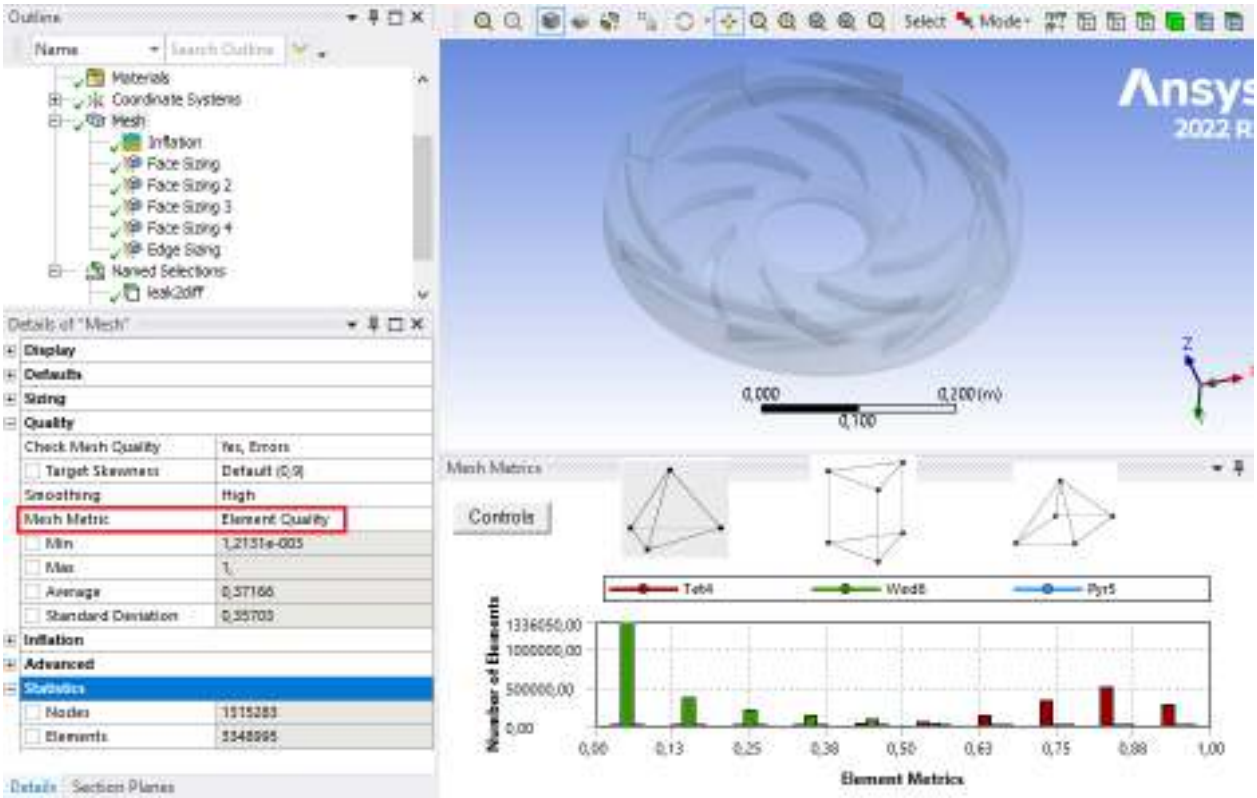


Figure 19: Check quality mesh

If the mesh does not meet the quality criteria it is necessary to solve the problem following different strategies:

- reducing the value of the global "Element Size".
- acting locally on the densifications with the "Sizing Edge/Faces"
- modifying the "Defeature Size"; there is no precise rule, this value can be increased as long as the geometry is not altered, in general it almost never exceeds 2mm.

This mesh generation process is iterative; we act on these parameters until a sufficiently accurate mesh is found that satisfies the mentioned quality criterion.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

MESHING LEAKAGE

1.1. Open the environment ANSYS WORKBENCH, drag and drop the two block "GEOMETRY" and "ICEM CFD" in the working area and link them as in Figure 20, then click "Import Geometry" for import the file leakage.iges inside the first block in the section geometry and open ICEMCFD.

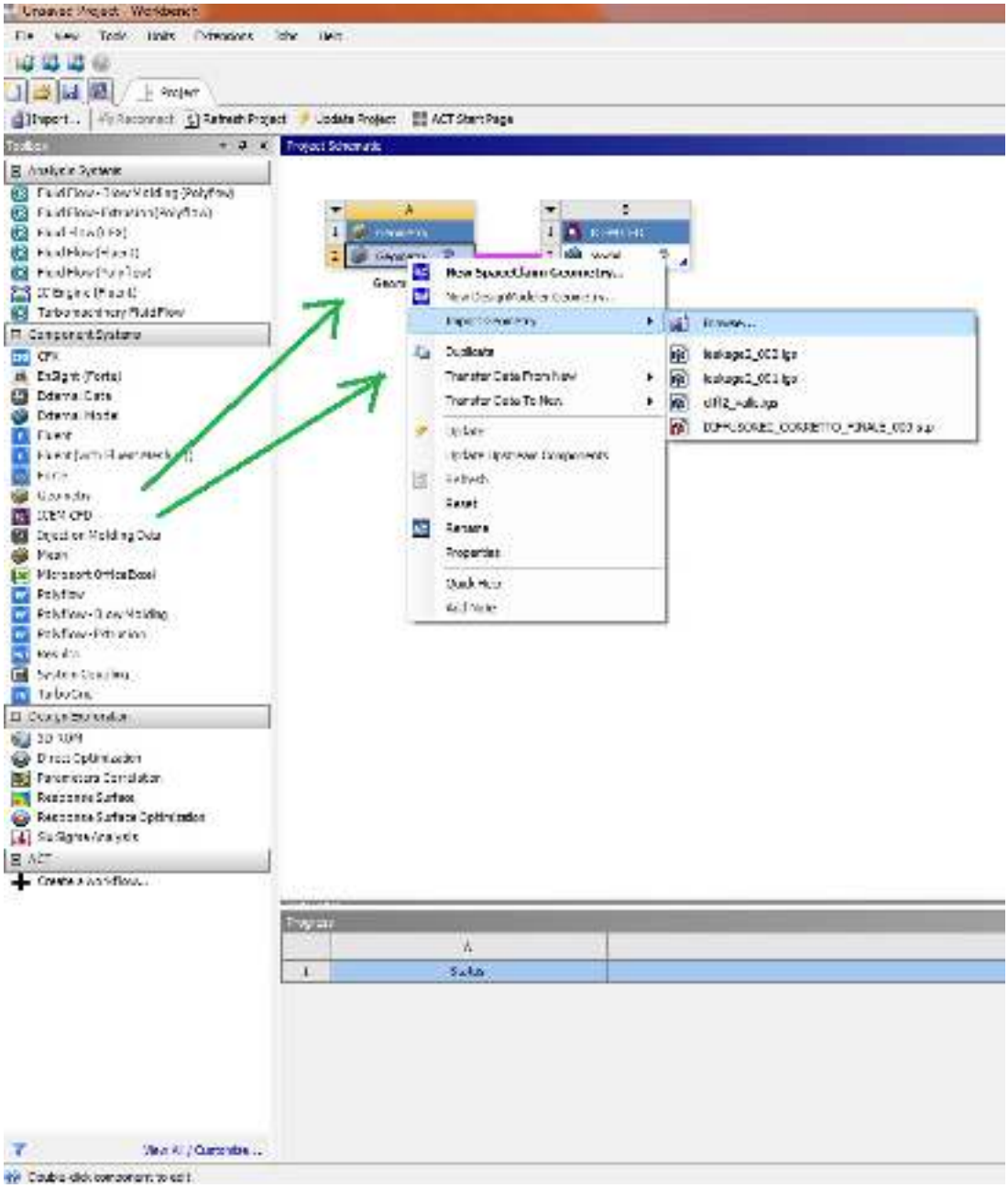


Figure 20 : working area of ANSYS WORKBENCH

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1.2. Create Block

Create the main block of the mesh.

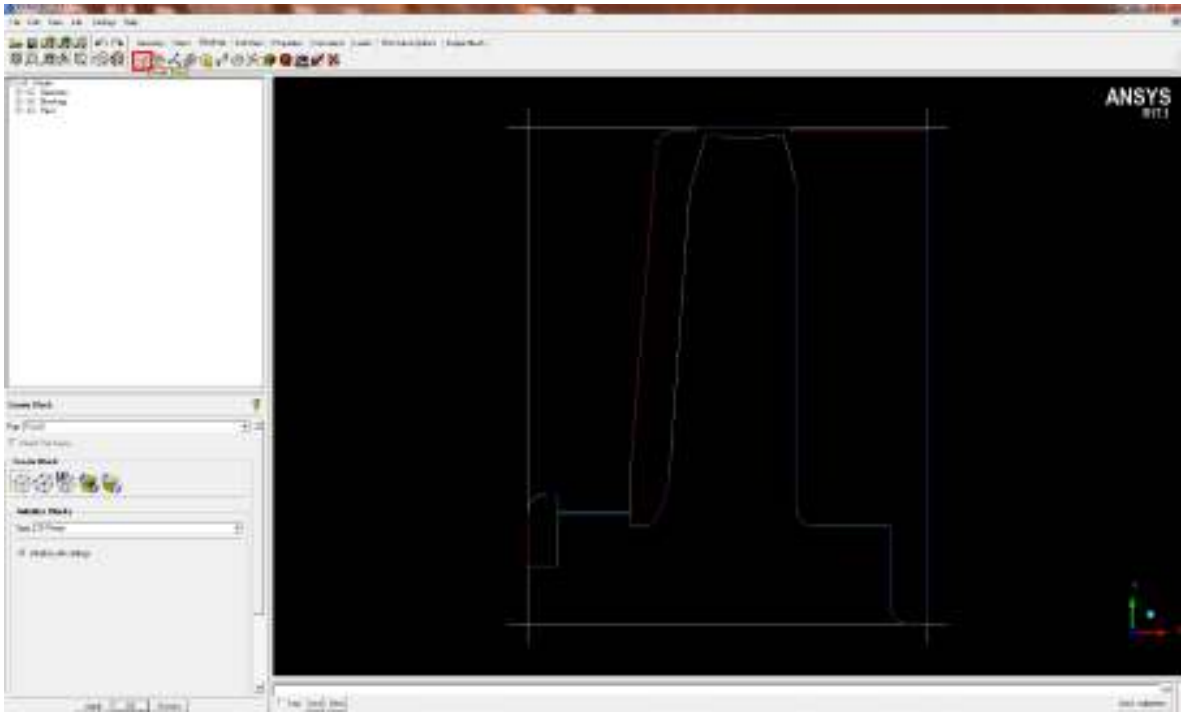


Figure 21 : Create block.

1.3. Split Block

Split the main block of the mesh into several blocks that follow the contour of the leakage.

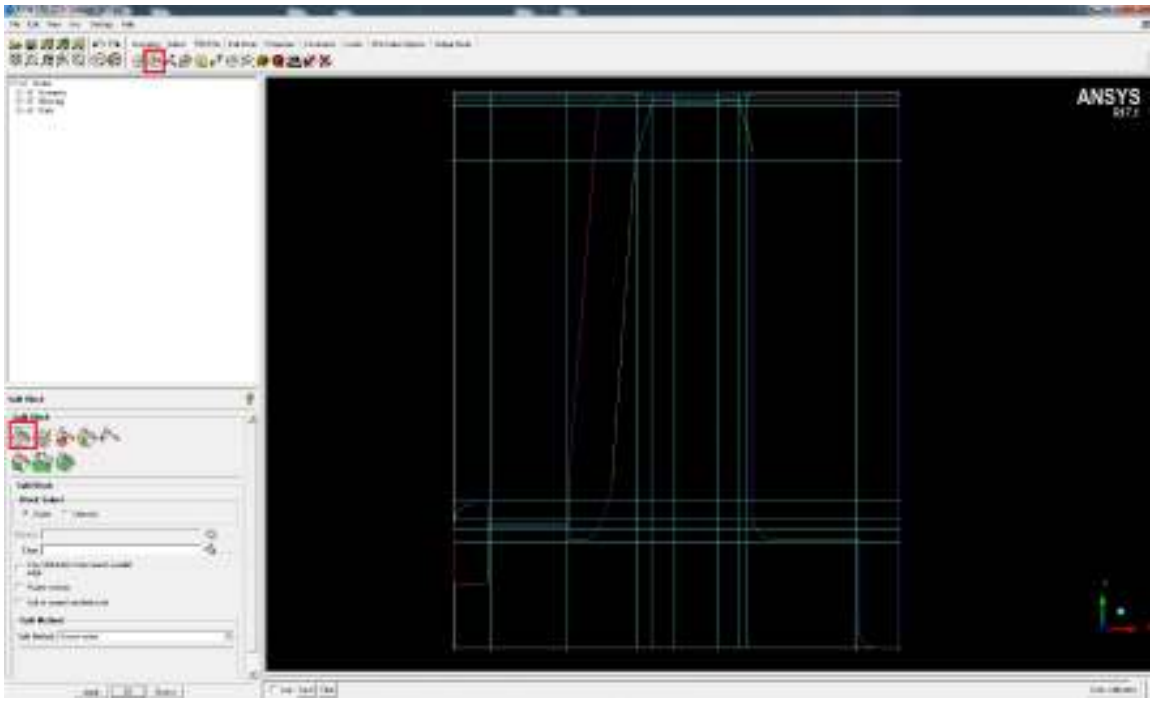


Figure 22 : Split block

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1.4. Delete Block

Delete excess blocks.

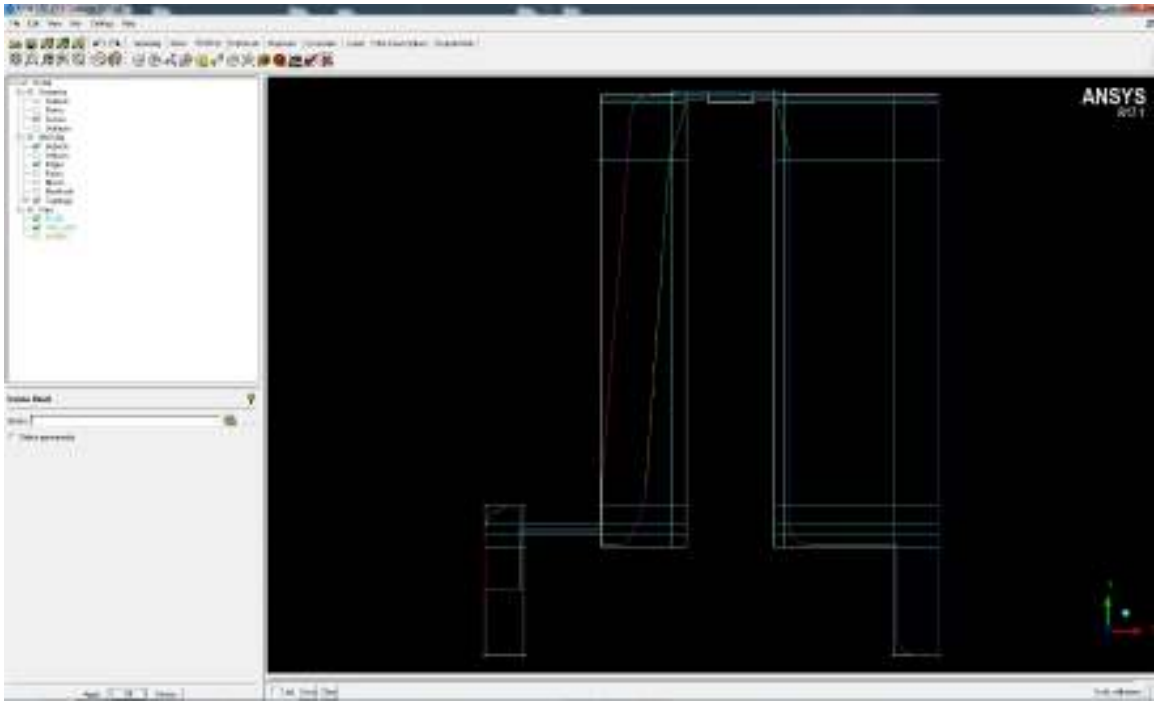


Figure 23 : Delete block.

1.5. "Associate Edge to Curve" and "Move Vertex"

Aassociate the edge of mesh with the curve of the geometry and move the vertex to adjust the orthogonality of the blocks.

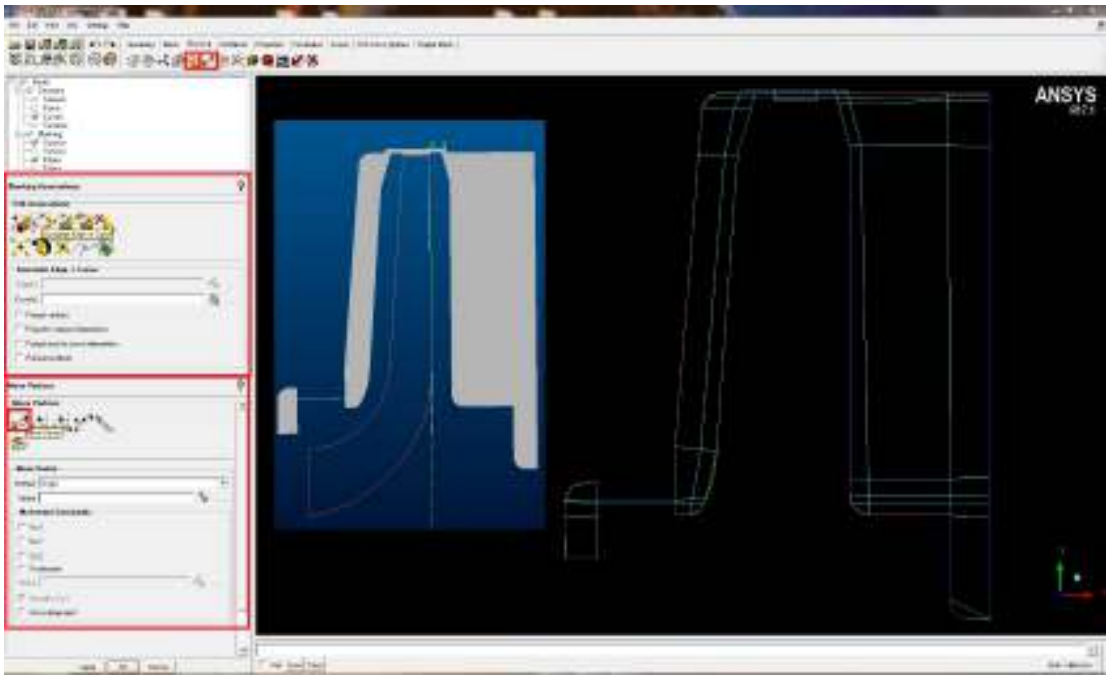


Figure 24 : "Associate Edge to Curve" and "Move Vertex

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1.6. Generate the grid mesh.

Select the edges and define the number of nodes for each block (use also “Copy Parameters”).

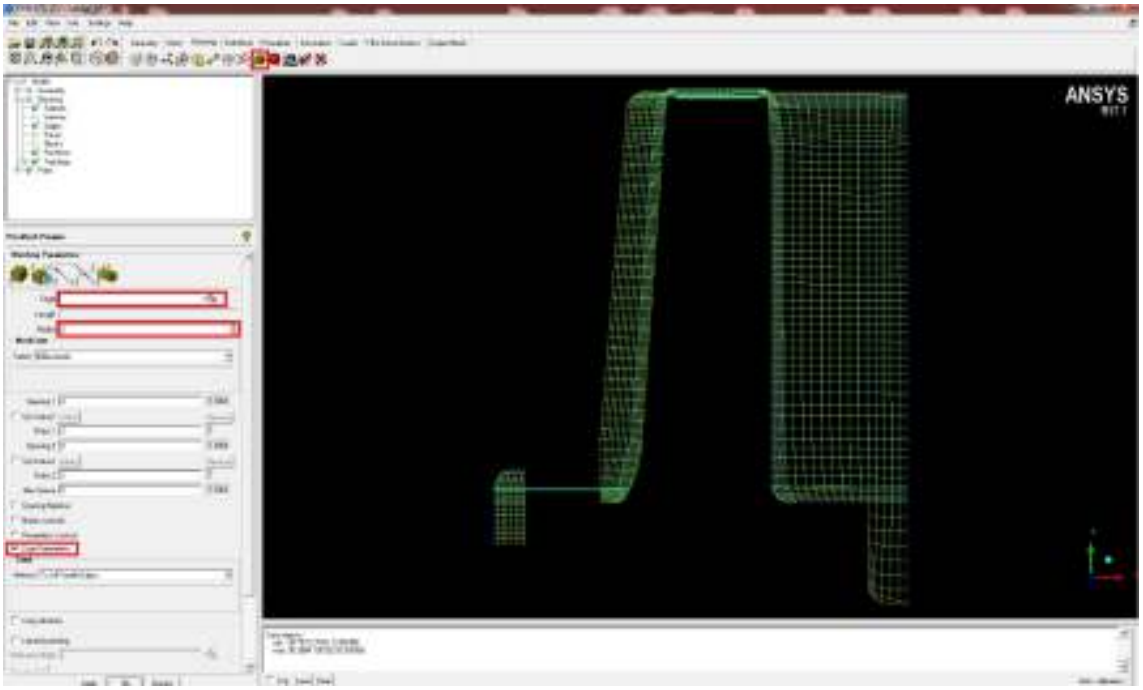


Figure 25 : Generate the grid mesh.

1.7. Generate “O grid”

Create the “O grid” for generate the boundary layers, deselect the edges interfaces.

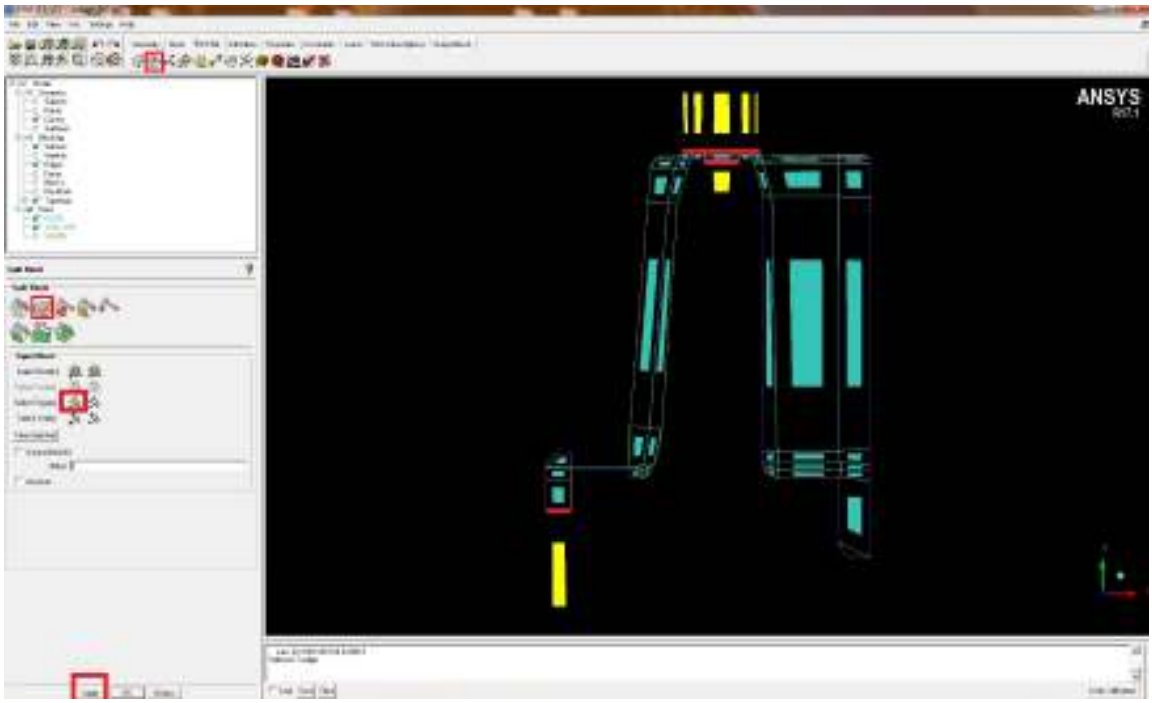


Figure 26 : Generate “O grid”

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1.8. Generate Boundary Layers

Create the Boundary Layers in “O grid” mesh, with 10 nodes, 1.4 of grow ratio and the thickness of first cell of 1micrometer.

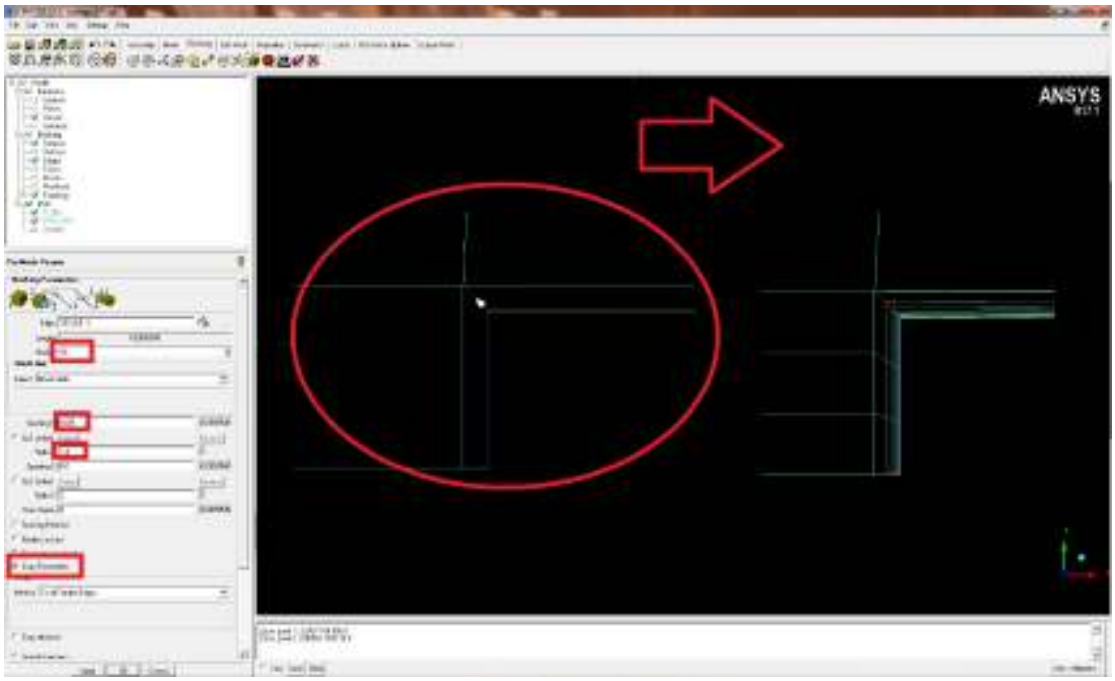
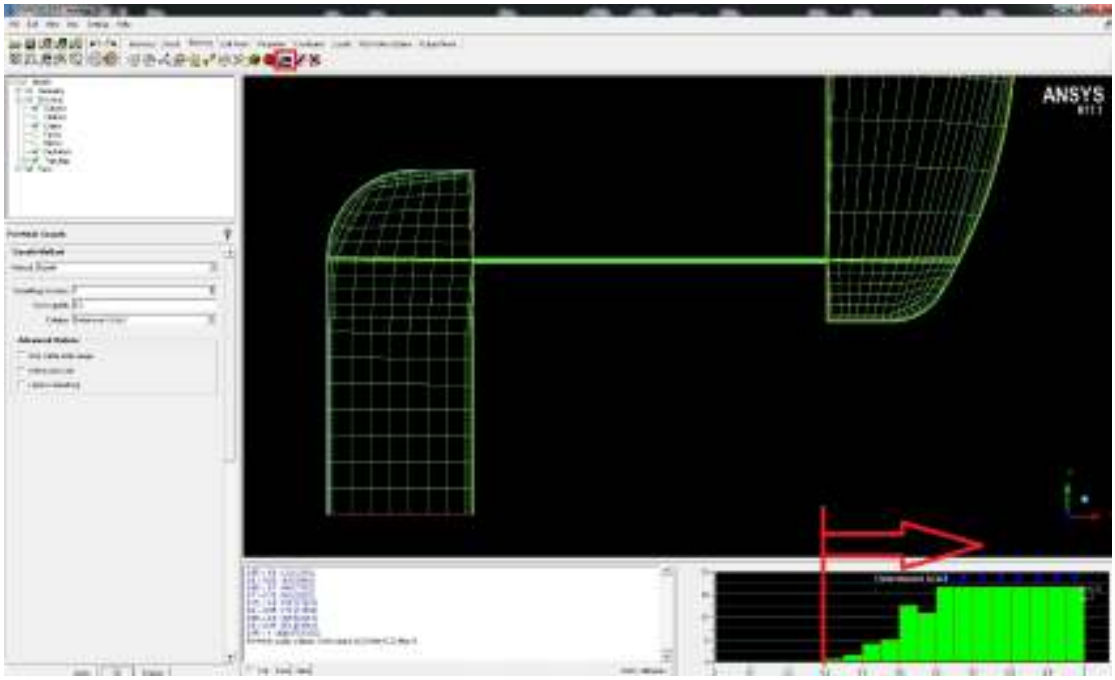


Figure 27 : Generate Boundary Layers

1.9. Check Quality of Mesh

Check Quality of mesh, verify that all element of mesh has been quality over 0.3.



COPY COMPLIANT to the original document resident on the computer system at the time of printing

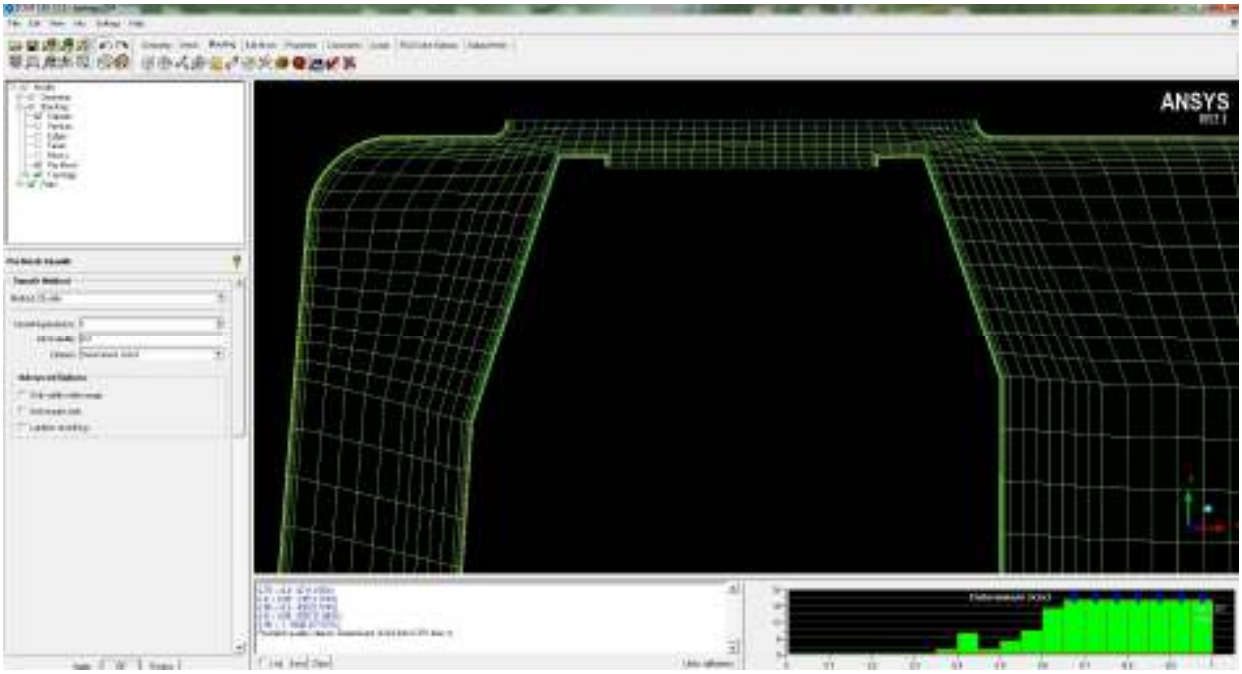


Figure 28 : Check Quality of Mesh

1.10. Convert to Unstruct Mesh

Convert the Struct Mesh to Unstruct Mesh

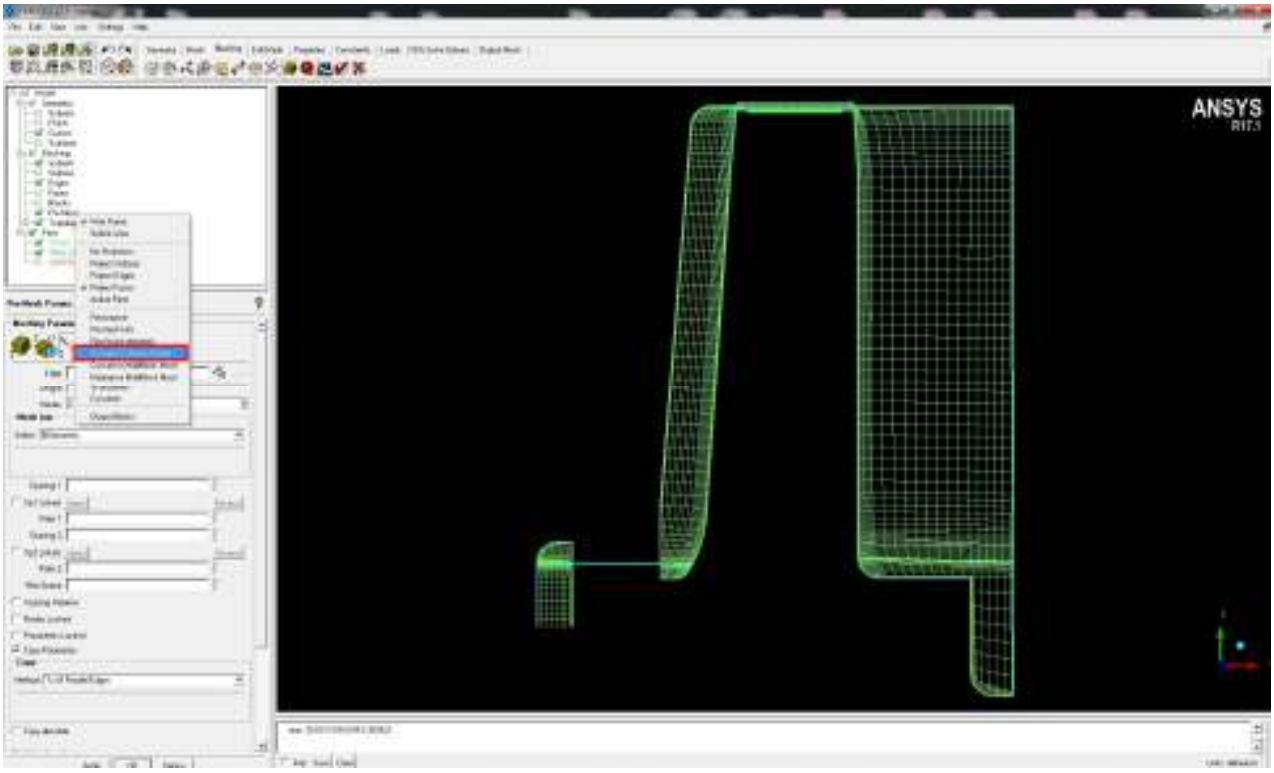


Figure 29 : Convert to Unstruct Mesh

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1.11. Extrude by Rotation Mesh

Select all mesh, create the revolve mesh respect the axis of rotation and divide the mesh rotation with 2degree for element in all complete rotation of 360degree.

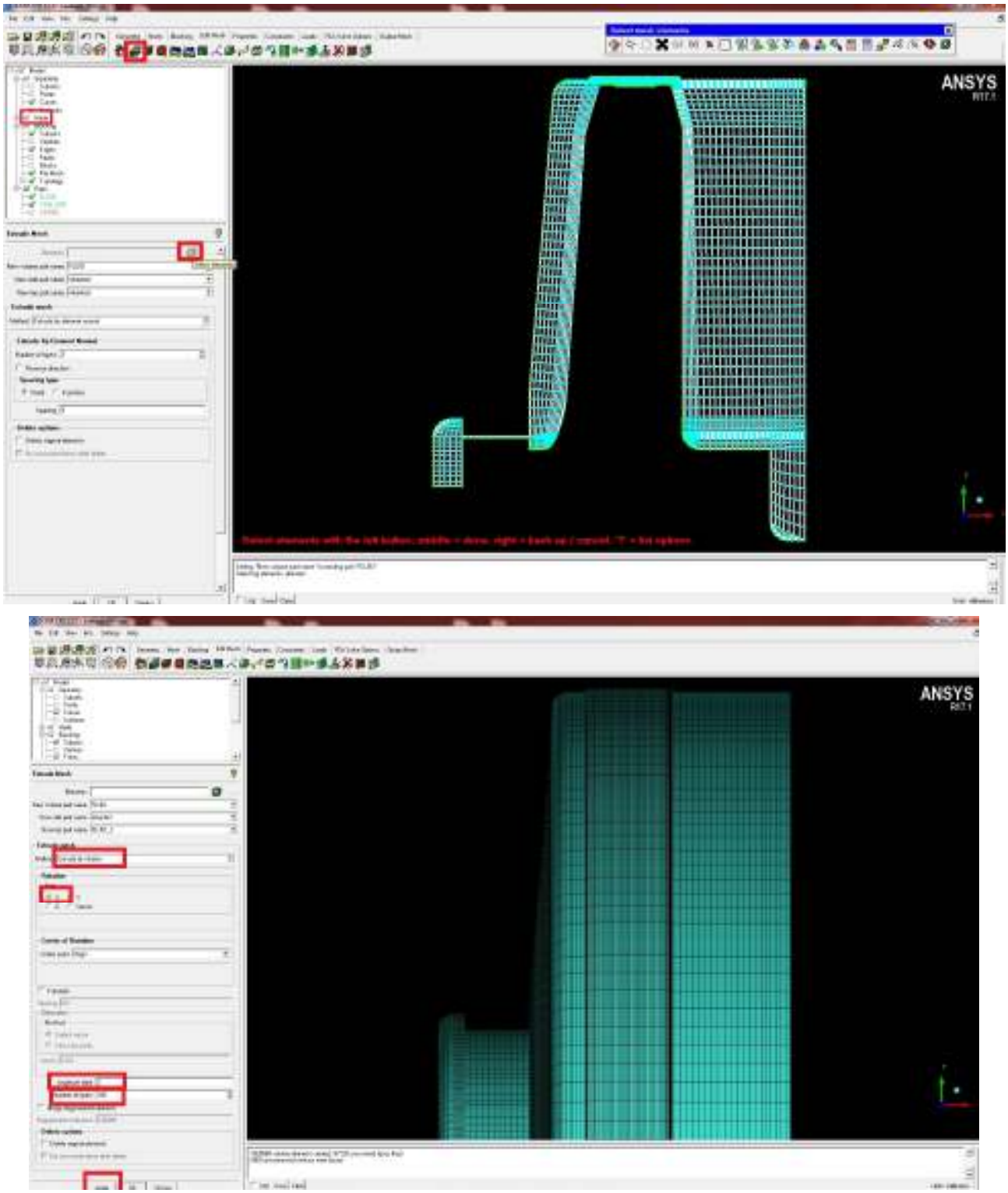


Figure 30 : Extrude by Rotation Mesh

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1.12. Create Part

Create the appropriate name for every patch, using the tool selection.

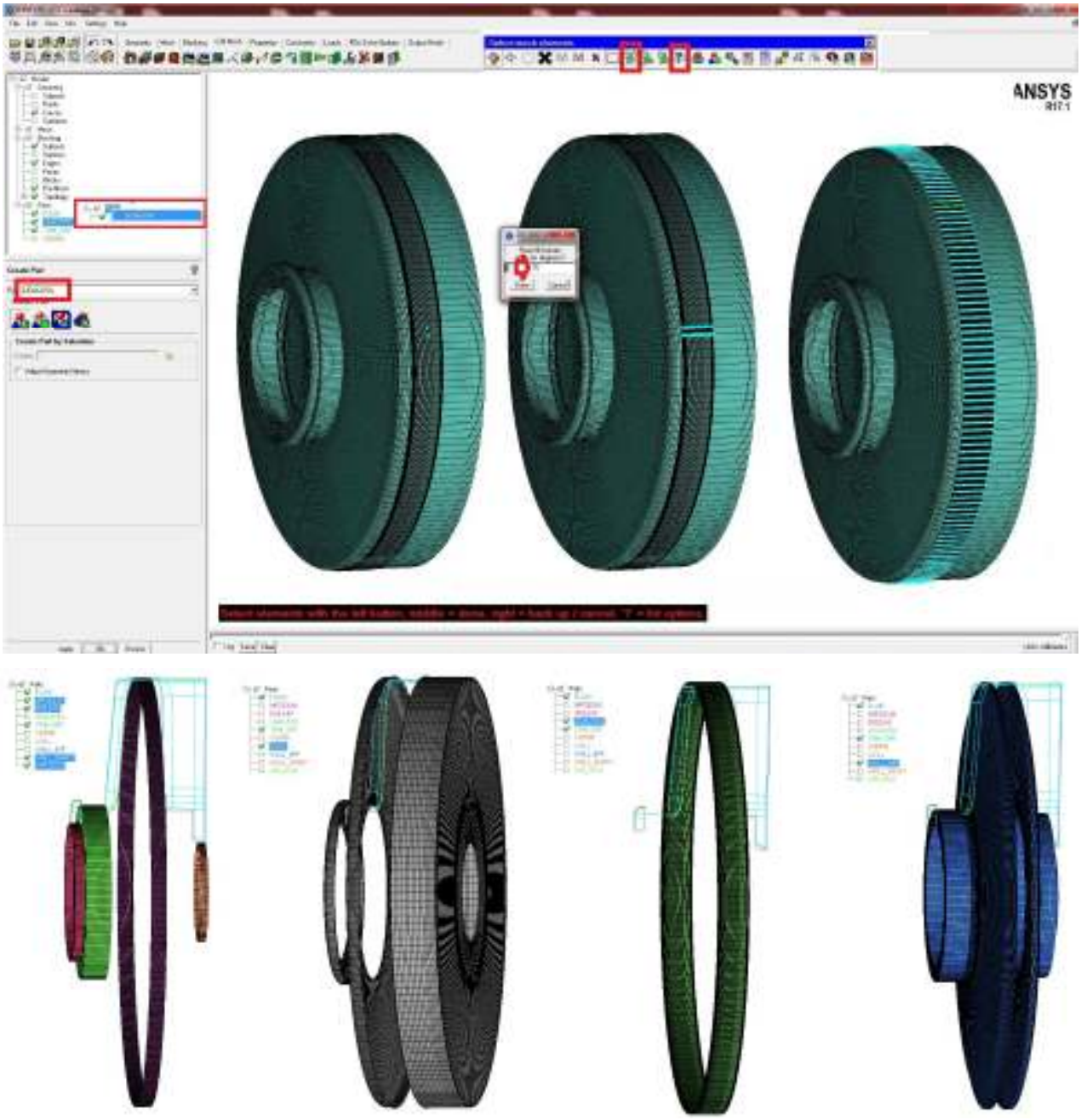


Figure 31 : Create Part

In this case of leakage has been use the follow parts:

- IMP2LEAK: is interface between domain impeller and domain leakage.
- IN2LEAK: is interface between domain suction and domain leakage.
- LEAK2VOL: is interface between domain leakage and domain volute.
- WALL: solid surface of the casing.
- WALL_IMP: solid surface of the impeller.
- WALL_SHAFT: solid surface of the shaft.
- WR_STAT: solid surface of the static wear ring.

COPY COMPLIANT to the original document resident on the computer system at the time of printing

1.13. Save mesh.

Select Output Solver (ANSYS CFX for example) and save the mesh.

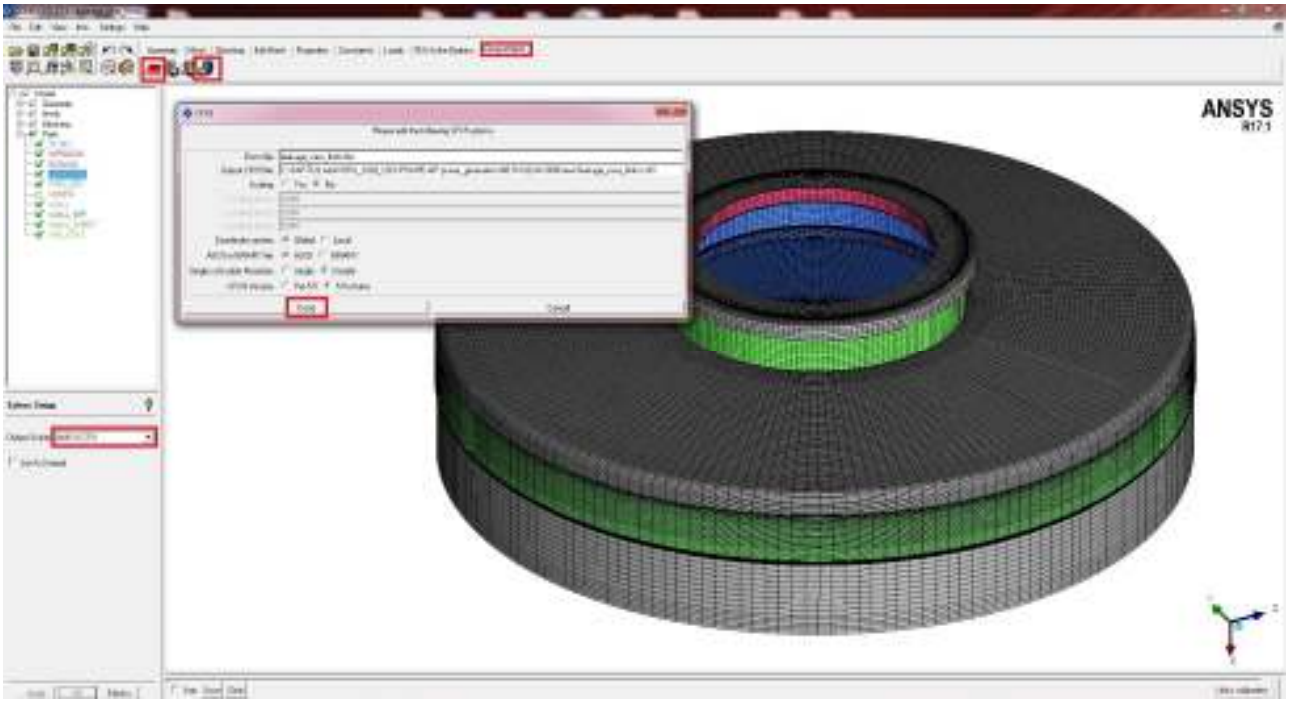


Figure 32 : Save mesh.

COPY COMPLIANT to the original document resident on the computer system at the time of printing